

REVIEW

Advancements in wearable heart sounds devices for the monitoring of cardiovascular diseases

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Abstract

Cardiovascular diseases remain a leading global cause of mortality, underscoring the urgent need for intelligent diagnostic tools to enhance early detection, prediction, diagnosis, prevention, treatment, and recovery. This demand has spurred the advancement of wearable and flexible technologies, revolutionizing continuous, noninvasive, and remote heart sound (HS) monitoring—a vital avenue for assessing heart activity. The conventional stethoscope, used to listen to HSs, has limitations in terms of its physical structure, as it is inflexible and bulky, which restricts its prospective applications. Recently, mechanoacoustic sensors have made remarkable advancements, evolving from primitive forms to soft, flexible, and wearable designs. This article provides an in-depth review of the latest scientific and technological advancements by addressing various topics, including different types of sensors, sensing materials, design principles, denoising techniques, and clinical applications of flexible and wearable HS sensors. This transformative potential lies in the capacity for ongoing, remote, and personalized monitoring, promising enhanced patient outcomes, amplified remote monitoring capabilities, and timely diagnoses. Last, the article highlights current challenges and prospects for the future, suggesting techniques to advance HS sensing technologies for exciting real-time applications.

KEYWORDS

cardiovascular diseases, clinical applications, denoising, heart sound monitoring, wearable technology

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1 | INTRODUCTION

Cardiovascular diseases (CVDs) include heart and blood vessel disorders, such as coronary artery disease, atrial fibrillation, heart failure, and hypertension.¹ Unhealthy lifestyle choices like physical inactivity, poor diet, smoking, and excessive alcohol consumption, along with underlying medical conditions like diabetes and obesity, contribute to the risk factors associated with CVDs. The substantial global impact of CVDs is evident in the high mortality rate of approximately 17.9 million people annually.² To mitigate the effects of CVDs, a comprehensive approach involving prevention, early detection, and effective treatment is essential. This includes promoting healthy lifestyles, conducting risk factor assessments, and ensuring access to quality healthcare services. Advancements in research and technology are crucial for developing innovative monitoring tools and improving patient outcomes while addressing healthcare disparities is vital to ensure equitable access to resources for all populations.

An essential approach to improve patient care involves monitoring and diagnosing heart conditions using various cardiovascular signals. These signals provide specific information about the heart's function, offering a comprehensive view of cardiovascular health. While the electrocardiogram (ECG) is the most common cardiovascular signal, it has limitations, particularly in diagnosing heart valve-related issues. Another valuable source of information about heart valve health and anomalies, such as heart murmurs, comes from the mechanoacoustic signals of the heart, known as phonocardiogram (PCG) and seismocardiogram (SCG). PCG is a noninvasive device that, combined with other cardiovascular signals, provides a more comprehensive heart assessment by recording heart sounds (HSs) and murmurs in an auditory mode.³ On the other hand, SCG, a noninvasive technique, measures HSs through chest vibrations caused by the heart's acoustic pressure during the cardiac cycle.⁴ Throughout this article, we shall refer to PCG and SCG as HSs since they were previously referred to as HSs in the medical literature.

During the cardiac cycle, the heart produces four fundamental sounds known as S1, S2, S3, and S4. S1 occurs when the heart contracts during systole to pump blood, while S2 occurs when the heart relaxes at the end of the systole to fill with blood. These sounds are crucial indicators of the heart's performance and can reveal the presence of any abnormalities.⁵ Heart murmurs and abnormal noises heard during the cardiac cycle may signal issues with the heart valves or other components. Monitoring HSs makes detecting abnormalities early and diagnosing CVDs such as valvular heart disease (VHD) possible.⁶

The conventional stethoscope, widely used for HSs analysis, has certain limitations that cannot be overlooked. Firstly, high selective sensitivity is required to compensate for the low amplitude of the combined high and low-frequency acoustic signal produced by HSs. Secondly, noise often distorts HSs signal obtained with a stethoscope, adversely affecting the accurate diagnosis of CVDs. Moreover, physicians assessing HSs with a stethoscope must possess extensive training, expertise, and keen hearing abilities for practical evaluation.^{4,7} Therefore, there is a demand for cost-effective, user-friendly, flexible, durable, and wearable technology to monitor HSs for CVD diagnosis. To address the limitations of the traditional stethoscope, wearable electronic stethoscopes employing different types of sensors, such as accelerometer, microphone, inertial measurement unit (IMU), force sensor, and micro-electromechanical system (MEMS) acoustic sensor, have been proposed. While commercially available rigid sensors reduce flexibility and wearability, researchers have developed sensors made entirely of polymer films, eliminating the need for rigid integrated circuits (ICs). These flexible and easy-to-wear sensors offer patients an overall improved monitoring experience.

HSs are generated at low frequencies (20–250 Hz),⁸ necessitating high sensitivity in devices to detect and analyze them accurately. In IC chip-based sensors (e.g., accelerometers, IMUs, force sensors, microphones) and MEMS acoustic sensors, the sensor type, structure, frequency range, and overall patch design (rigid or soft) influence the device sensitivity, specificity, and response time. Conversely, in polymer-based sensors, appropriate material selection (considering flexibility, conductivity, and biocompatibility), material structure (solid, porous, nanofiber, or nanotubes), device structure (sandwich or cantilever), and working mechanism (piezoelectric, piezoresistive, triboelectric, or capacitive) can impact the device overall performance. Figure 1 provides an overview of this review highlighting rigid and flexible wearable HSs devices with key features that affect device functionality and diseases that can be diagnosed using HSs.

Driven by the persistent prevalence of CVDs as a leading global cause of mortality, this comprehensive review aims to dissect the current advancements in wearable hyperspectral HSs monitoring systems and explore the prevailing technological challenges. Delving into crucial aspects of these monitoring devices, including their design principles, technological progressions, and clinical applications, particularly in CVD monitoring, the review will also confront the hurdles accompanying these technologies, such as the imperative for heightened sensitivity, mobility, robust materials, and

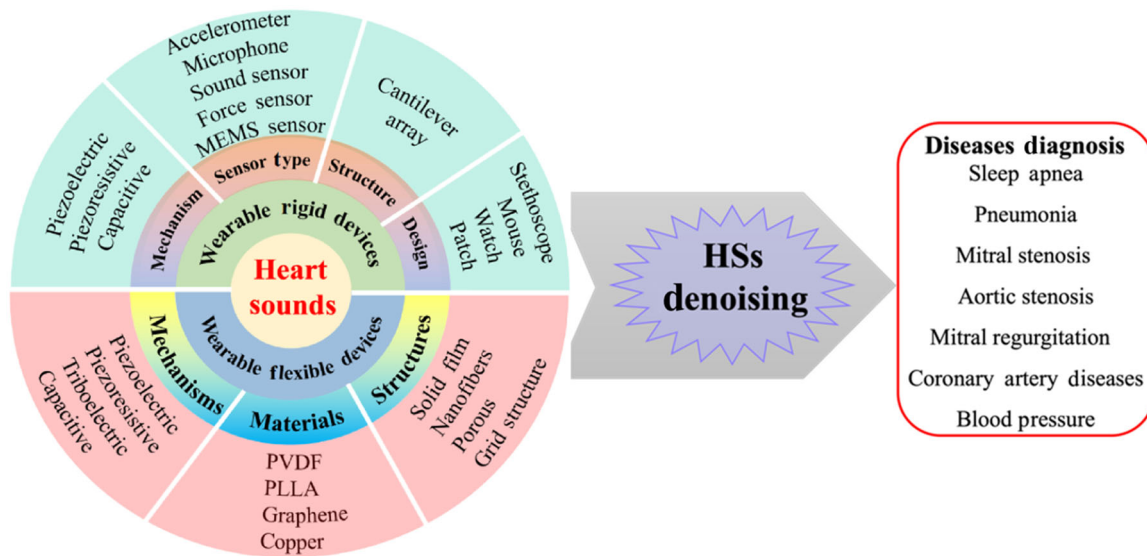


FIGURE 1 Overview of different types of HSs measuring devices for the diagnosis of diseases. HS, heart sound.

sophisticated denoising techniques. By meticulously addressing these challenges, this review intends to lay the groundwork for further research and clinical adoption of wearable HSs monitoring technologies. Offering invaluable insights into future trajectories and prospects for innovation in this domain, the review endeavors to meticulously scrutinize the barriers and constraints of these technologies, thereby elucidating potential avenues for advancement and refinement.

2 | HEART SOUNDS

2.1 | Understanding aberrant HSs

The human heart comprises four chambers: two atria and two ventricles, as Figure 2A depicts. These chambers are responsible for receiving and pumping blood throughout the body, facilitated by heart valves. The heart valves open and close passively in response to the surrounding blood's inertial forces. There are two main types of heart valves: atrioventricular (tricuspid and mitral) and semilunar (pulmonary and aortic) valves (Figure 2A).⁹ These valves permit blood flow into and out of the heart during specific phases known as diastole and systole, respectively, as illustrated in Figure 2B. One complete diastole and systole phase constitutes a cardiac cycle (Figure 2B). Consequently, the opening and closing of these heart valves produce HSs.¹⁰ Different HSs are generated during the diastole and systole phases, and in the following section, we will provide a detailed description and classification of these HSs. Additionally, various aberrant HSs exist alongside the normal HSs,

some of which are harmless, while others may indicate more significant cardiac issues.

2.2 | Classification of HSs

The human heart's rhythmic diastole and systole phases give rise to four discernible HSs, categorized as the first HS (S1), second HS (S2), third HS (S3), and fourth HS (S4), as demonstrated in Figure 2C.¹¹

2.2.1 | Primary HSs

The primary HSs perceptible within the cardiac cycle are denoted as S1 and S2, colloquially known as “lub” and “dub,” owing to their distinctive auditory characteristics, as depicted in Figure 2B.¹² These sounds emerge from the closure of the atrioventricular and semilunar valves, occurring sequentially with each pulse.¹² In brief, S1 corresponds to the cessation of the movement of the tricuspid and mitral valves. While both valves contribute to this sound, the dominant source is the mitral valve due to its louder closure, attributed to elevated pressure within the left side of the heart. S1 persists for approximately 0.15 s, oscillating between 25 and 45 Hz frequencies.^{8,11}

Conversely, the culmination of ventricular systole leads to the closure of the aortic and pulmonic valves, instigating the S2 sound, which heralds the onset of diastole. The closure of the left aortic valve predominantly shapes the S2 HS due to heightened pressure on the heart's left side. With a duration of 0.12 s and a frequency of 50 Hz, S2 encapsulates the intricate transition

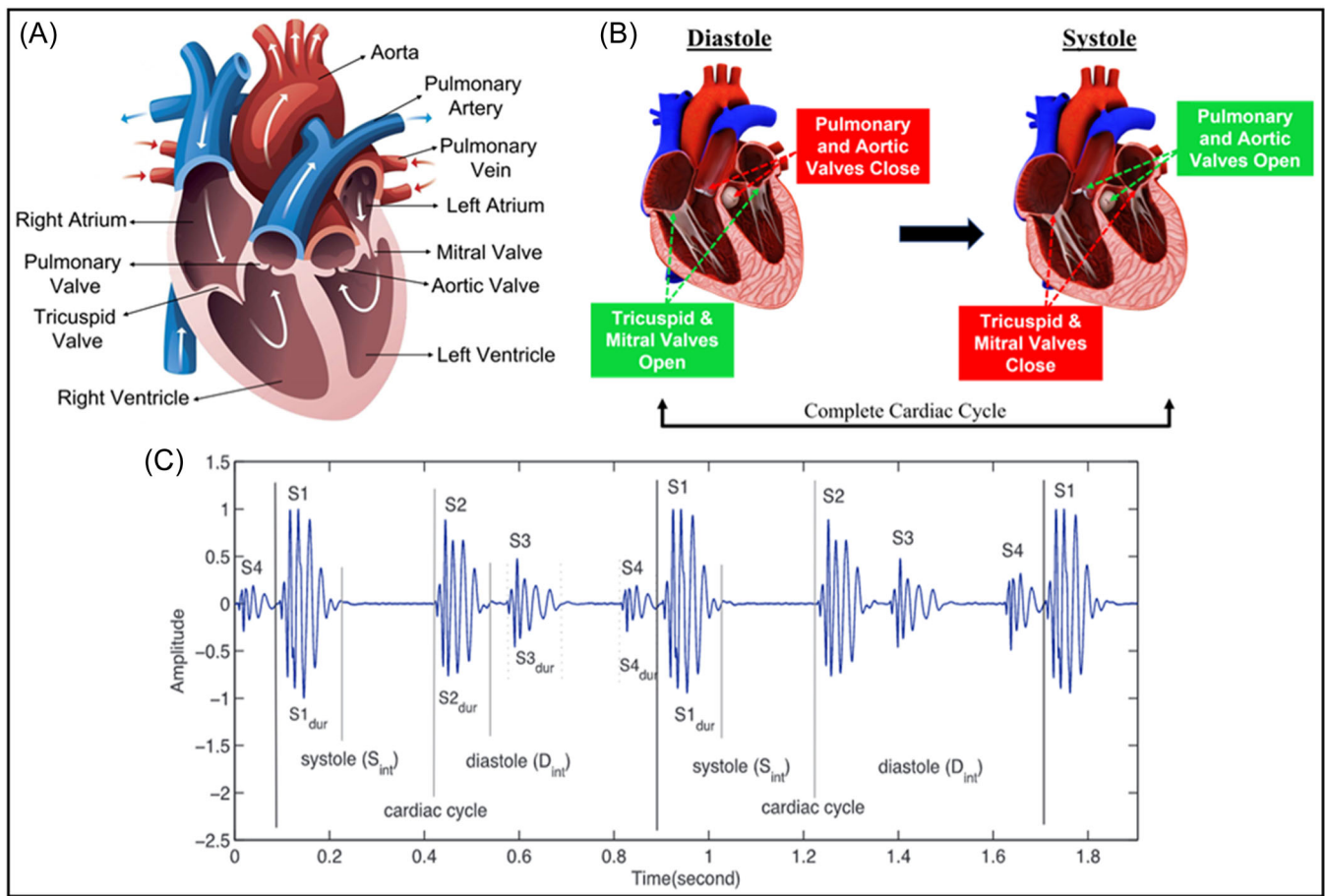


FIGURE 2 Heart anatomy and graphical representation of HSs. (A) Description of human heart parts. (B) Representation of a complete cardiac cycle, i.e., diastole and systole phase. (C) Pattern of S1, S2, S3, and S4 (Reproduced under the terms of CC-BY-NC-ND license: Copyright 2022, Elsevier.¹⁰). HS, heart sound.

between cardiac phases.^{8,11} Alterations in these HSs' frequency, intensity, or duration can reveal crucial insights into conditions encompassing malfunctioning heart valves, heart failure, and other cardiac ailments.

2.2.2 | Secondary HSs

In addition to the primary HSs, S1 and S2, two secondary HSs, S3 and S4, manifest during diastole and are intricately linked to ventricular filling. S3 emerges when the mitral valve permits a substantial influx of blood into the left ventricle. Although its audibility is sporadic, it can occasionally be perceived, particularly among young adults. S3 follows the S2 with a temporal lag of 0.1–0.2 s. Notably, the pitch of S2 often surpasses that of S3, as depicted in Figure 2C.

Contrastingly, S4 transpires in the late diastole before S1. It coincides with atrial contraction, propelling blood into the left ventricle. Notably, S4's distinctive low-pitched tone signifies a potential indicator of diastolic

heart failure, as illustrated in Figure 2C. The stethoscope bell proves most effective in optimally discerning this sound. While S1 and S2 are fundamental HSs, S3 and S4 introduce additional auditory elements. The latter, often called gallop sounds, may or may not be discernible in a healthy adult heart.^{11,13}

2.2.3 | Cardiac murmur

Heart murmur constitutes another type of cardiac sound that becomes discernible when there is an aberration in the usual blood flow within the heart. This gives rise to auditory phenomena encompassing sounds like blowing, rasping, whooshing, or swishing, originating from the heart or its immediate vicinity. Such murmurs manifest due to insufficient closure of heart valves or the presence of blood flowing across a constricted valve. This turbulence in blood circulation can be attributed to various factors, including valve anomalies, congenital heart irregularities, and myocardial abnormalities.

The characterization of murmurs is influenced by the phase of the cardiac cycle during which they are detected, leading to their classification as systolic, diastolic, or continuous murmurs. Systolic murmurs emerge between the first and second HSs, potentially indicating conditions like aortic stenosis, mitral regurgitation, or ventricular septal defects. On the other hand, Diastolic murmurs arise between the second and first HSs, with a notable trio of common heart disorders—specifically aortic regurgitation, mitral stenosis, and tricuspid regurgitation. In contrast, continuous murmurs present throughout the entire cardiac cycle, often arising from conditions such as patent ductus arteriosus and arteriovenous fistulas.

While faint murmurs may not necessarily signify heart pathology, robust murmurs or irregular acoustic fluctuations can indicate an underlying cardiac anomaly.¹⁴ The distinct origin of murmurs manifests at varied phases of the cardiac cycle, yielding frequencies spanning from 100 to 600 Hz.⁸ An outline of significant parameters characterizing both normal and abnormal HSs is illustrated in Figure 3.

3 | HSS MEASUREMENT TECHNOLOGIES

The detection of HSs is conventionally accomplished through a stethoscope. Since its inception by René Laennec in 1816, the stethoscope has remained an indispensable instrument in the medical field for diagnostic purposes.^{15,16} Although its core design has endured over time, advancements in materials, electronics, and manufacturing methodologies have influenced its contemporary appearance. Stethoscopes play a pivotal role in diagnosing and treating various medical conditions.^{17,18} Auscultation conducted via a stethoscope offers a noninvasive, expeditious, and information-rich diagnostic approach in scenarios where imaging is impractical or unavailable.

To conduct precise auscultation and provide patients with optimal care, the selection of an appropriate stethoscope is of paramount importance.¹⁹ A proficient stethoscope should be able to discern clinically significant sounds while concurrently attenuating extraneous noises, such as biological and environmental interferences, that lack clinical relevance. In situations necessitating sound amplification or the recording and transmission of auscultatory findings (as encountered in telemedicine applications), digital stethoscopes likely offer a superior option.^{7,20,21} This is owing to their capability to transmit and archive auscultated sounds.

When selecting a stethoscope, numerous individuals consider the chest piece material, tubing durability, and earpiece comfort. However, the decision is not always solely predicated on the acoustic attributes and performance of the instrument; at times, aesthetic considerations may come into play. Additionally, the analysis of recorded HSs for early CVD detection and cost-effectiveness are pivotal factors to consider. Given the constraints of presently available stethoscopes, a pressing need exists for advanced technologies capable of surmounting these limitations. Consequently, diverse methodologies have been employed to devise wearable sensors capable of capturing HSs by measuring chest wall vibrations in a straightforward and analytically tractable manner.

Among these methods, sensors leveraging IC chips, encompassing readily available accelerometers, microphones, IMUs, and MEMS, have garnered extensive utilization. Specific microwave and radar systems have also been proposed for HSs measurements. In parallel, explorations into polymer film-based flexible, lightweight, and wearable sensors for HSs have also been undertaken.

The selection of the appropriate sensing technology for HSs detection relies upon considering factors such as sensitivity, noise susceptibility, durability, cost-effectiveness, and physical form. The integration of these parameters guides the decision-making process. The ensuing section will concisely provide an overview of the varied sensing techniques investigated by researchers for the measurement of HSs.

3.1 | IC chip-based devices

Semiconductor IC chips are engineered to execute specific functions by integrating many distinct electronic components, including resistors, diodes, and transistors.²² Ongoing technological progress has facilitated the creation of substantial devices condensed onto a single chip, often measuring only a fraction of a millimeter or micrometer in scale. This breakthrough has spurred the production of a diverse array of IC chips tailored for measuring parameters such as acceleration, vibration, and force, with many of these chips readily available in the commercial market. Additionally, these chips have the capacity for continuous data recording. Consequently, considerable research has been directed toward fabricating various preassembled IC chips equipped with sensor modules to monitor HSs.

3.1.1 | Accelerometer

Choosing a suitable sensing method for HSs detection is a complex decision-making process governed by several

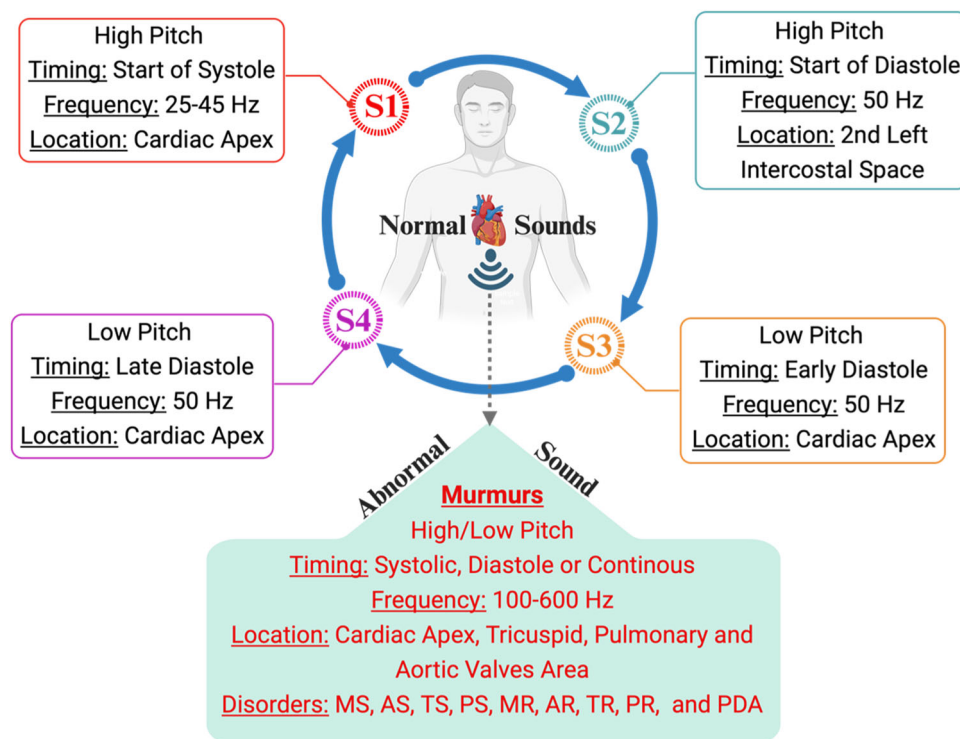


FIGURE 3 Classification of normal and abnormal HSs. HS, heart sound.

pivotal criteria. Among these is sensitivity, a crucial parameter considering that HSs are characterized by low-frequency acoustics, necessitating a sensor with outstanding sensitivity for precise detection. An alternative approach involves the deployment of accelerometers. Accelerometers are significant for monitoring physical activities because they can measure acceleration, which allows for monitoring human motion. The classification of actions, such as walking, can be achieved by monitoring the acceleration profile of an accelerometer attached to a body. Based on the mechanism, modern accelerometers are piezoelectric and capacitive accelerometers. But recently, capacitive accelerometers have gained widespread adoption as the preferred form of accelerometer due to their user-friendly nature, durability, and sensitivity. Capacitive accelerometers have been incorporated into many electronic products, including smartphones, cameras, and gaming systems. Capacitive accelerometers consist of a seismic mass suspended between fixed anchor “arms”, as depicted in Figure 4A(i). When in a state of balance/rest, the movable arms of the seismic mass are positioned at a distance d away from the stationary arms, as seen in Figure 4A(i). As the sensor undergoes acceleration, the position of the floating mass shifts, resulting in a change in the distance between the moveable and fixed arms [Figure 4A(ii)]. The change in distance leads to a proportional variation in capacitance, which can be utilized to compute acceleration.²³ These

instruments excel at capturing the vibrations propagated by the heart's acoustic waves, methodically monitoring alterations in acceleration along the x , y , and z axes. Acceleration is assessed by measuring mass displacement or the frequency of a vibrating object.²⁴

Employing an accelerometer on the human chest facilitates the discernment of minuscule vibrations, converting them into electrical signals contingent upon the specific structural arrangement, often involving capacitive effects. In comparison to other sensing methods, accelerometers demonstrate enhanced resistance to noise and electromagnetic interference. However, they may have lower sensitivity, requiring more sophisticated data processing techniques.

Additionally, a singular chip sound sensor can discern HSs and convert them into electrical impulses for assessment.²⁷ This specific sensor design incorporates an analog-to-digital converter (ADC), signal conditioning hardware, and an electromechanical sensor/sound transducer. The heart's acoustic signal necessitates initial transformation by the electromechanical sensor/sound transducer before generating an electrical representation. Subsequently, the analog electrical signal converts into a digital format, enabling processing by a computer or other digital device through the ADC. Last the signal conditioning circuitry filters and amplifies the digital signal, enhancing its accuracy and fidelity. Consequently, various models of accelerometers and

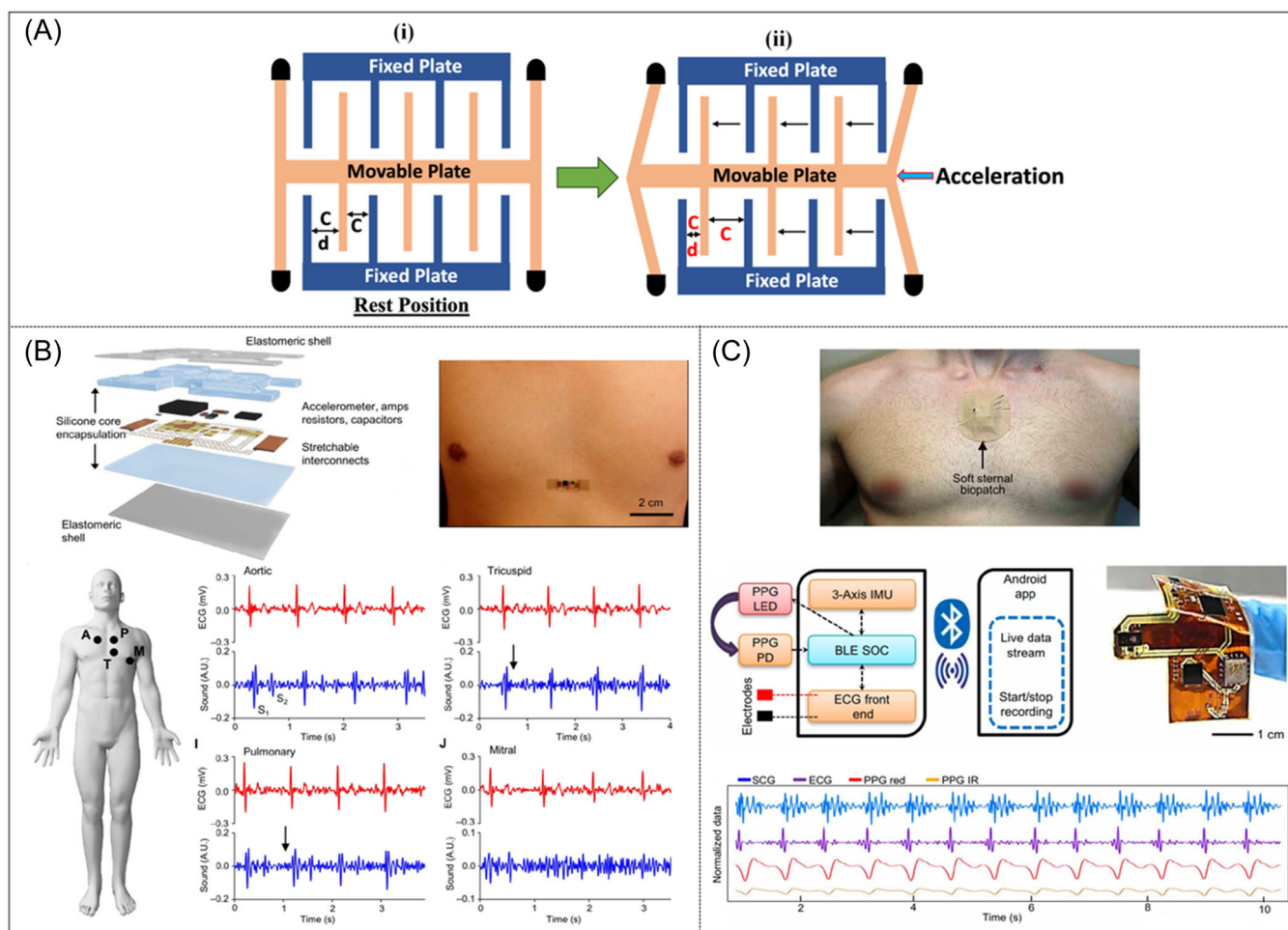


FIGURE 4 Accelerometers based wearable patches for the recording of HSs. (A) Accelerometer working principle. (B) Epidermal mechano-acoustic HSs sensor (Reproduced under the terms of CC-BY-NC 4 license: Copyright 2016, Science.²⁵). (C) A wireless soft patch-based HSs monitoring. (Reproduced under the terms of CC-BY 4 license: Copyright 2021, Science.²⁶). HS, heart sound.

sound sensors have been deployed to detect and quantify HS.

Wearable devices utilizing commercially available, cost-effective accelerometers such as BMA280,²⁸ MPU-9250,²⁹ and ADXL335²⁵ have been investigated for real-time monitoring of HSs. However, due to the rigid structure of the device, prolonged wear may lead to discomfort, and improper skin-device contact can compromise accuracy. Consequently, efforts have been directed towards developing stretchable interconnects, achieved by encapsulating a low-modulus silicone shell with an ultralow elastomeric core²⁵ (Figure 4B). These partially flexible interconnectors facilitate secure attachment to the skin, effectively eliminating air gaps and mitigating potential noise or distortion. Furthermore, combining a three-axis accelerometer and gyroscope, the IMU has been leveraged for HSs measurement through chest vibrations.^{26,30} Based on IMU technology, Zavanelli et al.²⁶ introduced a

wireless soft patch utilizing HSs to detect sleep apnea (Figure 4C).

3.1.2 | Microphones

The detection of HSs constitutes a pivotal phase within the cardiological diagnostic process, offering indispensable insights into cardiac function and overall health status. As previously elucidated, the dominant sensors utilized for HSs detection predominantly encompass acoustic sensors such as microphones. Notably, MEMS-based microphones have gained prominence as acoustic sensors for HSs assessment. Among these, piezoelectric, piezoresistive, and capacitive microphones are three critical categories grounded in distinct operational principles.

The piezoelectric MEMS microphone harnesses the piezoelectric effect, whereby pressure, force, or stress triggers charge generation across the attached

piezoelectric material layers' top and bottom electrode sheets.³¹ Figure 5A illustrates the comprehensive structural design of the piezoelectric microphone. Following the impact of HSs vibrations, the diaphragm experiences compression and relaxation, thereby inducing deformation in the piezoelectric material layer. This deformation culminates in generating electrical charges via electrodes, constituting the sensor's output.³²

The Capacitive MEMS microphone operates as a variable capacitor, featuring two silicon plates: stationary and movable. Perforations are incorporated into the fixed plate electrode, allowing sound transmission while maintaining an electrical coupling that facilitates signal reception. The interaction of HSs occurs through these apertures, inducing a change in capacitance between the stationary and movable plates due to the deformation of the latter prompted by sound pressure.³¹ Subsequently, the IC within the microphone assembly calculates the alteration in capacitance and translates it into an electrical signal. The operational process of the MEMS capacitive microphone is graphically represented in Figure 5B.

Various types of microphones and sound sensors have been examined for the development of electronic stethoscopes aimed at detecting HSs.^{6,37–39} Lee et al.^{40,41} employed a MEMS microphone and microphone model ICS40212 to fabricate a soft wearable stethoscope and a soft patch for continuous cardiac biometric monitoring, respectively. These patches, enhanced with machine learning algorithms and a Bluetooth module, enable wireless diagnosis of diverse lung diseases and biometric locking, achieving 95% and 99.55% accuracy rates, respectively. Similarly, using microphones (ICS-43434) and (POM-2783L-Lw100-R), a computer mouse-shaped stethoscope (Figure 5C),³³ and the tablet-based wearable patch has been proposed for the monitoring of HSs.⁴² In addition to microphones, a continuous cardiac auscultation monitoring system based on the sound sensor HKY-06B has been presented by Ren et al.⁴³ This device excels in HSs acquisition, storage, and asynchronous analysis, achieving a detection accuracy of 86.66%.

The optimal placement of a single-sensor device on the chest is a crucial consideration owing to the distinct locations of the four heart valves, which directly impact the quality of HSs signal capture. Consequently, defining the most suitable positioning for such a device presents a significant challenge. To address this issue, various approaches have been reported in the literature to enhance the accurate monitoring of HSs. These include using array structures comprising sound sensors^{34,35} (Figure 5D,E), microphone,⁴⁴ and adhesive stick designs.⁴⁵

Among the notable advantages of capacitive MEMS microphones are their compact dimensions, heightened sensitivity to low-frequency sounds, and minimal power consumption. Capacitive models are not exempt from certain limitations like MEMS microphone variants. One such drawback pertains to potential accuracy deviations in readings due to external factors like temperature and humidity. Additionally, their dynamic range, indicating the span of sound pressure levels accurately detectable, also poses a constraint. Notably, the low-frequency range makes them particularly susceptible to environmental noise interference. Capacitive MEMS microphones stand as a burgeoning technology, offering superior sensitivity and energy efficiency in HSs detection. This advancement has the potential to refine the precision and reliability of cardiac diagnostics while enabling the development of innovative wearable and implantable cardiovascular monitoring solutions.

In addition to accelerometers, microphones, and sound sensors, alternative technologies such as force sensors (utilizing force cardiography Figure 5F),^{36,46} fiber optic technology,⁴⁷ noncontact radar-based methodologies,^{48,49} self-injection-locked oscillator with tag antenna,⁵⁰ and microwave techniques^{51,52} have been explored for HSs monitoring. Each operates on distinct principles, and this diversity of technologies underscores the range of approaches to detecting and monitoring HSs.

IC-based sensors exhibit certain advantages that confer them superiority in HSs measurement. Their diminutive size, low power consumption, exceptional sensitivity, and ability to discern low-frequency sounds render them highly suitable for incorporation into wearable devices for continuous HSs monitoring. Furthermore, they can be integrated with other sensor systems and wireless communication channels, facilitating remote data transmission and real-time monitoring. These attributes hold the potential to elevate the precision and reliability of cardiac diagnostics, offering a foundation for the development of advanced wearable and implanted cardiovascular monitoring systems. In conclusion, IC-based sensors have the potential to enhance the accuracy and reliability of cardiac diagnostic methodologies significantly.

Nonetheless, there are noteworthy limitations that warrant consideration. Despite being fabricated on flexible substrates, there exists a trade-off between flexibility and the structural integrity of the device. Consequently, prolonged wear may lead to discomfort and a restriction in mobility. Furthermore, the rigid structure of these devices may lead to challenges in achieving proper contact with the skin, potentially impacting sensitivity levels. Additionally, it is imperative to ensure a stable power source for IC-based sensors, which may pose challenges

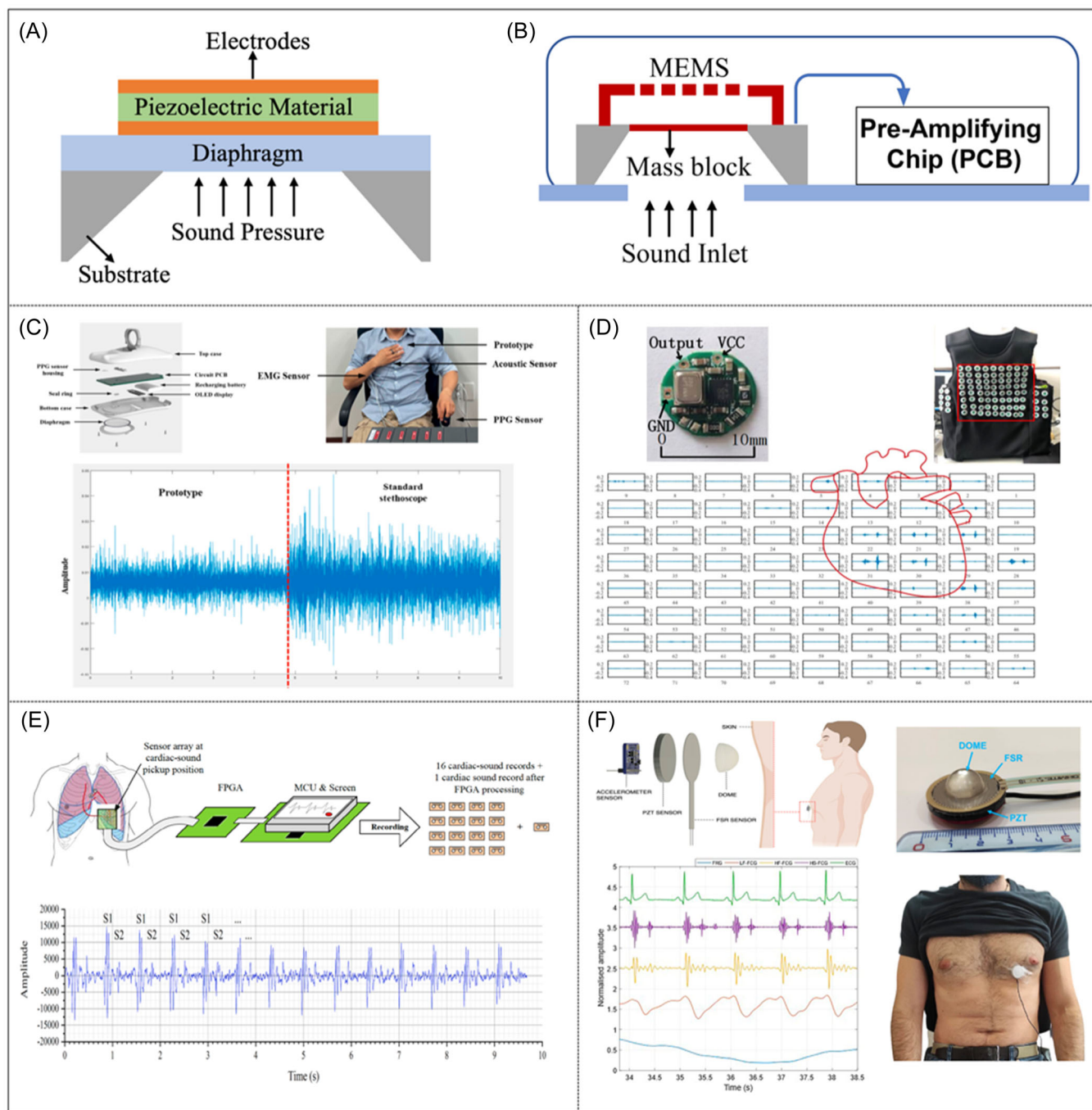


FIGURE 5 Microphones-based wearable HSs devices. Working principle of (A) Piezoelectric and (B) Piezoresistive microphones. (C) Computer mouse shaped stethoscope (Reproduced under the terms of CC-BY 4 license: Copyright 2022, MDPI.³³). (D and E) Array structured of microphone and sound sensor based HSs device (Reproduced under the terms of CC-BY 4 license: Copyright 2022, MDPI.³⁴ and Reproduced under the terms of CC-BY-NC-ND license: Copyright 2021, Elsevier,³⁵ respectively). (F) Force sensor-based HSs monitoring (Reproduced under the terms of CC-BY 4 license: Copyright 2021, Frontiers.³⁶). HS, heart sound.

in certain applications due to the specific power requirements of these sensors.

Notwithstanding these limitations, researchers have substantially advanced in developing electronic stethoscopes and wearable devices that employ IC chip-based sensors for HSs detection. A comprehensive

listing of IC chip-based wearable devices/stethoscopes designed for HSs measurement is provided in Table 1. Selecting the most appropriate sensor hinges upon the specific prerequisites the application dictates, as each sensor exhibits its unique strengths and weaknesses.

3.2 | MEMS-based HSs devices

MEMS constitute diminutive integrated systems or devices that amalgamate mechanical and electrical components. Depending on the specific requirement, their dimensions can vary from a few micrometers to a few centimeters.⁵⁶ The fabrication of MEMS entails a sequence of pivotal procedures, encompassing wafer cleaning, oxidation, resistance bar etching, low boron ion implantation, deoxidization, high boron ion etching, sputtering, and material release.^{57,58} Compared to conventional production methodologies, MEMS technology, presents several advantages, notably heightened reproducibility, precision, and the capacity to craft intricate devices featuring substantial aspect ratios.⁵⁹

Researchers have introduced highly sensitive HSs sensors featuring diverse structural designs, leveraging the microfabrication capabilities of MEMS. For instance, Li et al.⁶⁰ employed MEMS technology to develop a bionic hydrophone tailored for HSs detection (Figure 6A). Inspired by the neuromast organ found in fish, this hydrophone capitalizes on the ciliary motion induced by the impinging HSs. Consequently, the beams experience oscillatory movement, leading to variations in the resistance of the piezoresistors affixed to them following the piezoresistive mechanism. This hydrophone exhibits the capacity for continuous HSs monitoring, characterized by a notably heightened sensitivity. It attains an impressive 8.2 dB superior signal-to-noise ratio (SNR) compared to the 3200-type 3 M Littmann stethoscope. Similarly, Duan et al.⁶¹ and Shi et al.⁶² developed electronic stethoscopes with the same bionic MEMS structure. However, these devices exhibit lower sensitivity, with SNR measuring 2.2 and 2.59 dB higher than the 3 M Littmann 3200 electronic stethoscope when compared to.⁶⁰ The MEMS HSs sensor was developed with a T-type sensitive microstructure comprising two main components: a bionic cilium and a central mass-cantilever. When HSs signals interact with the bionic cilium, it causes a deviation in the bionic cilium. This deviation leads to a change in fatigue on the cantilever beam, resulting in changes in the resistance of the piezoresistors. The resistance changes of piezoresistors are typically not easily detected by signal conditioning circuits. These changes are converted to voltage changes that can be detected by a Wheatstone bridge. Thus detection of HSs signals is achieved by surpassing the SNR of 3200-type stethoscopes by 7.9 dB.⁶³ Additionally, concave, racket-shaped cilium⁶⁴-based (Figure 6B) HSs sensors have been reported that replicate the structural mechanism of 3D fiber bundles found in the otic hair cells of humans. This microstructure consists of cilium and high-precision cantilever beams,

incorporating the bionic principle, piezoresistive effect, and MEMS technology. The cilium base is securely attached to the center junction of the beam components. At the same time, varistors with the same resistance are positioned beneath the area of highest stress in the cantilever beams, creating a Wheatstone bridge. When sound waves are directed towards the biomimetic cilium, they undergo oscillation, causing deformation in the cantilever beams. Ultimately, the bending of the cantilever leads to an alteration in the electrical resistance of the varistor positioned on the surface of the cantilever. The Wheatstone bridge converts the mechanical deformation into an electrical voltage output. Thus, this structure achieved high sensitivity for the recording of HSs with SNR value of 27.05 dB.

Gupta et al.⁶⁵ introduced a torsional cantilever structure designed to serve as a vibration sensor for cardiopulmonary signals, as depicted in Figure 6C. This sensor operates by translating micro-gravity level accelerations resulting from chest wall vibrations into torque, thereby inducing a change in capacitance. This alteration can be detected at the electrodes, demonstrating sensitivity across a broad bandwidth (DC-12 kHz). Additionally, a silicone-based cantilever structure has been explored as a piezoresistive HSs sensor, leveraging changes in internal resistance due to chest vibrations. This design exhibits superior performance at a lower cost compared to the 3200-type electronic stethoscope.^{66,67}

In addition to piezoresistive approaches, Qu et al.,⁶⁸ Jia et al.,⁶⁹ and Ning et al.⁷⁰ fabricated wearable MEMS acoustic sensors based on piezoelectric aluminum nitride (AlN) arrays for HSs monitoring. These devices operate on the principle of the piezoelectric effect. When these sensors come into contact with human skin, the sounds produced by the heart travel through tissues, blood, and skin, causing the thin sensing diaphragms to stretch. The stretching of the sensor diaphragms results in perpendicular strains, which, in turn, produce electric charges through the piezoelectric effect. The array structure facilitates the alteration of acoustic signals to electrical signals and enables the monitoring of cardiopulmonary signals from the human body. The proposed sensors yield SNRs of 13.58 and 52 dB at 500 Hz, respectively, comparable to or even surpassing those of commercial electronic stethoscopes. This attests to their ability to detect HSs with high sensitivity and precision. An overview of MEMS-based acoustic sensors for potential HSs monitoring applications is provided in Table 2.

MEMS-based sensors offer distinct advantages over conventional HSs detection counterparts. They exhibit high sensitivity, possess a compact physical profile, and can be fabricated on flexible substrates, rendering them

TABLE 1 Summary of IC chip-based wearable devices to monitor the HSs.

Sensor type	Nature of device	Sensor location	Applications	Reference
3-axis Acc.	Partially soft and flexible	Four heart valves	Cardiovascular diagnostic & speech recognition	[25]
3-axis Acc.	Rigid	Chest	Hemodynamics variables	[28]
3-axis Acc.	Partially soft and flexible	Sternum	Sleep apnea	[26]
Accelerometer	Rigid	Tricuspid valve	Cardiac health monitoring	[53]
9-axis IMU	Rigid	Sternum	Heartbeat detection	[29]
Microphone	Rigid	Chest	Cardiovascular health monitoring	[33]
Microphone	Partially flexible	Chest	Cardiac sounds	[40]
Sound Sensor	Rigid	Chest	Cardiac auscultation	[43]
Microphone	Rigid	Chest	HSs visualization	[34]
Microphone	Partially soft and flexible	Chest	HSs	[41]
Microphone	Rigid	Chest	Heart and lung sounds	[42]
Microphone	Rigid	Chest	HSs	[44]
Microphone	Rigid	Chest	Cardiopulmonary sounds	[45]
Microphone	Rigid	Chest	HSs	[54]
FSR-Acc.	Rigid	Xiphoid process	Heart mechanical vibrations	[46]
Force Sensor	Rigid	Chest	Respiration and HSs	[36]
ECM	Rigid	Mitral area	Cardiac sounds	[35]
Stethoscope	Rigid	Sternum	Estimation of ECG, PEP, LVET, and respiration rate	[55]

Abbreviations: Acc., accelerometer; ECM, electret-condenser-microphone; FSR, force sensitive resistor; HS, heart sound; IMU, inertial measurement unit; IC, integrated circuits.

apt for integration into wearable devices tailored for continuous HSs monitoring. The amalgamation of these sensors with various electronic components and signal-processing techniques has facilitated the development of innovative diagnostic tools and wearable and implantable cardiovascular monitoring devices that surpass prior accuracy, sensitivity, and cost-effectiveness methods.

However, their practical implementation in health-care is constrained by intricate and time-consuming fabrication processes and their rigid structure. Furthermore, a critical limitation of the devices mentioned above (IC and MEMS-based) lies in the constrained scope of experimental trials, predominantly involving healthy subjects. This impedes the ability to diagnose heart diseases reliably. To ascertain accuracy, reliability, and stability, rigorous testing on diverse populations, encompassing individuals with various heart conditions, ages, and genders, is imperative.

Validation of the accuracy and reliability of IC chip and MEMS-based HSs sensors for clinical applications necessitates substantial clinical research. Such studies unearth potential limitations or challenges associated with these sensors, thereby informing the development of refined diagnostic tools and advanced wearable/implantable cardiovascular monitoring systems that offer heightened precision, sensitivity, and dependability.

3.3 | Polymers-based flexible HSs sensors

Using soft and pliable epidermal electronics for HSs sensing presents several distinct advantages compared to conventional rigid sensors. Soft, flexible sensors can conform to the skin's surface, thus minimizing the air gap between the sensor and the skin. This adjustment

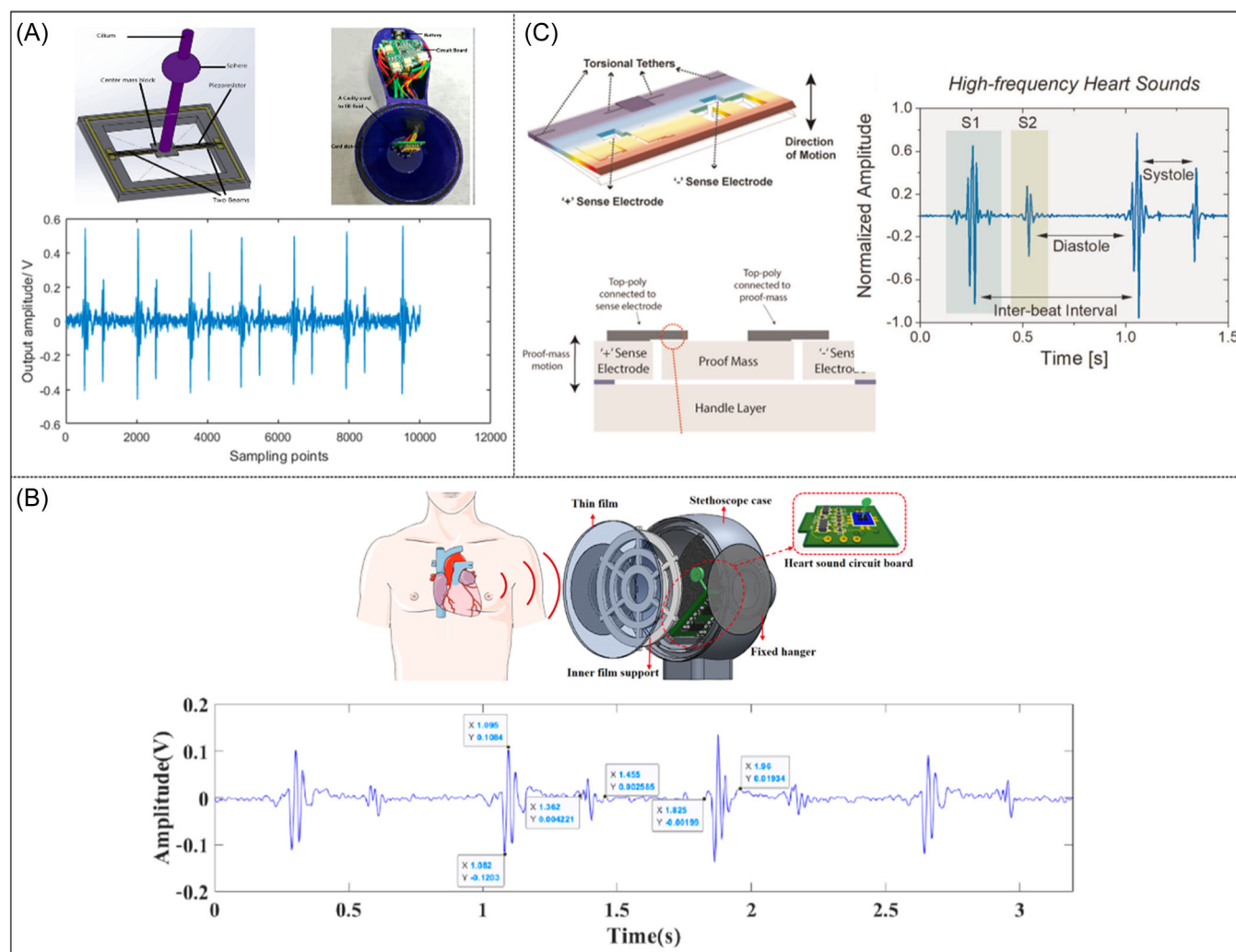


FIGURE 6 MEMS-based electronic stethoscopes. (A) MEMS bionic hydrophone (Reproduced under the terms of CC-BY 4 license: Copyright 2019, AIP.⁶⁰). (B) Concave cilium MEMS HSs sensor (Reproduced under the terms of CC-BY 4 license: Copyright 2021, MDPI.⁶⁴). (C) Cantilever structured accelerometer contact microphone (Reproduced under the terms of CC-BY 4 license: Copyright 2021, Nature.⁶⁵). HS, heart sound.

leads to heightened sensitivity and improved acoustic coupling.⁷² Furthermore, due to their inherent biocompatible properties, soft sensors can be worn for extended durations without causing discomfort or irritation to the wearer.

Various polymers have recently been explored in developing soft, lightweight, and flexible HSs sensors due to their flexibility, stretchability, and biocompatibility. Given the low frequency of HSs, the choice of sensing materials and their structural configuration are crucial factors in designing highly sensitive HSs sensors to ensure optimal performance and comfort.⁷³ The advancement of soft and flexible HSs sensors, utilizing polymers and flexible electronic circuitry, represents a promising avenue for continuously monitoring cardiac function. This section will delve into potential sensing materials and working

mechanisms for fabricating wearable and flexible devices for HSs monitoring.

3.3.1 | Working principles

Specific flexible polymer films have demonstrated the capacity to detect HSs by undergoing structural deformation in response to chest vibrations. Depending on the material's characteristics, this deformation induces piezoelectric, piezoresistive, triboelectric, or capacitive phenomena within the films. Among these mechanisms, the piezoelectric principle stands out for its widespread application in HSs detection, attributed to its high sensitivity. At the same time, other mechanisms have also been explored for measuring HSs. Figure 7A illustrates the operational principle based on the piezoelectric

TABLE 2 Summary of MEMS-based wearable HSs devices.

Structure	Nature of device	Sensitivity	Applications	Reference
Bionic hydrophone	Rigid	SNR 15.10 dB	HSs	[60]
Fish's neuromast	Rigid	SNR 41.40 dB	HSs	[61]
Cantilever	Rigid	SNR −189.50 dB @500 Hz	HSs	[62]
T-type	Rigid	SNR −183 dB @500 Hz	Diagnosis of CVDs	[63]
Concave Cilium	Rigid	SNR −180.60 dB @500 Hz	HSs	[64]
Cantilever	Rigid	76 mV/g	Cardiopulmonary signals	[65]
Cantilever	Rigid	SNR 27 dB	HSs	[66]
Cantilever	Rigid	SNR 58 dB	Health monitoring	[67]
6 by 7 array	Rigid	SNR 13.58 dB	Physiological sounds	[68]
PMUT array	Rigid	−154 dB ± 0.50 dB (re: 1 V/μPa)	Internet of Medical Things (IoMT)	[69]
Triangular cantilever	Rigid	SNR 17 dB	HSs	[70]
8 × 9 Array	Rigid	SNR 52 dB	Cardiopulmonary signals	[71]

Abbreviations: HS, heart sound; PMUT, piezoelectric micromachined ultrasonic transducer; SNR, signal to noise ratio.

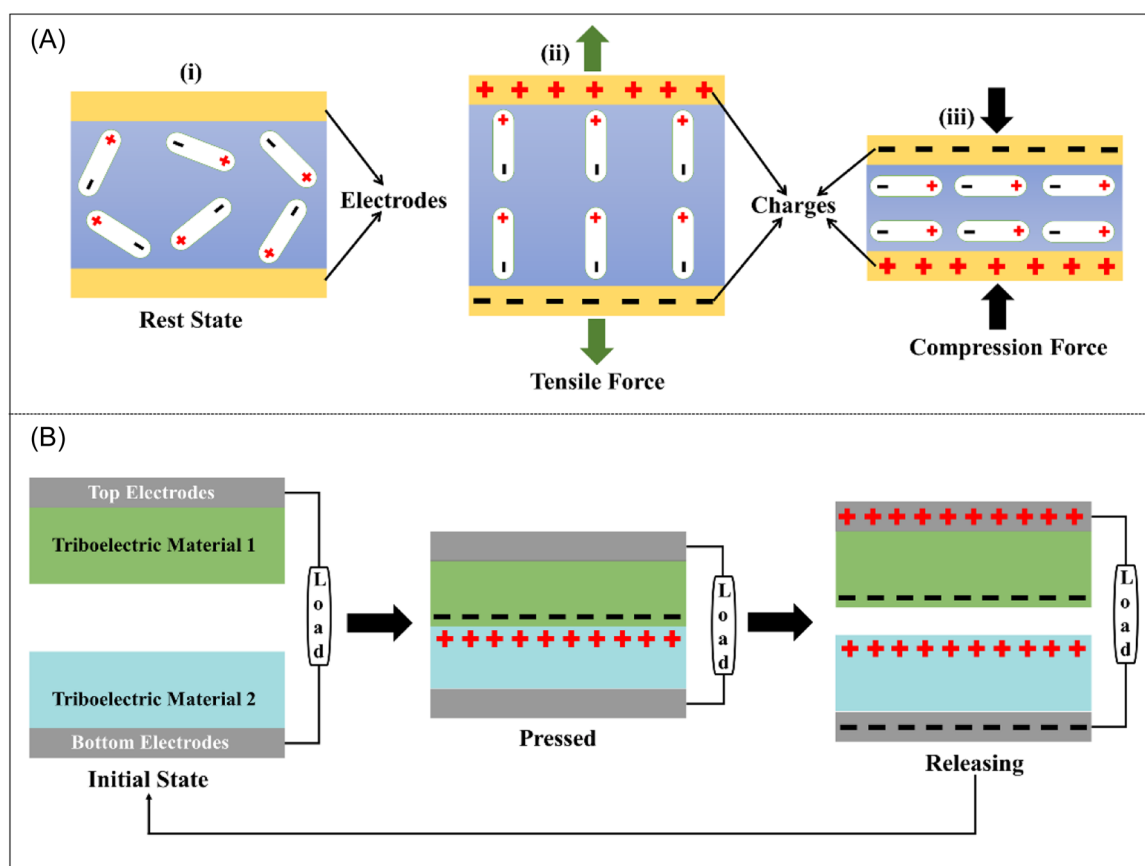


FIGURE 7 Flexible sensors working mechanisms (A) Piezoelectric effect (B) Triboelectric principle.

effect. The piezoelectric effect is the charge generation process and voltage on both sides of a material.⁷⁴ When an external force (tensile or compression force) is applied to a piezoelectric material in a specific direction, the

formation of electric polarization within the material occurs. Simultaneously, electric charges of positive and negative signs are produced on the two surfaces of the crystal. The material reverts to a neutral state when the

external force is removed. The alignment of the electric charge's polarity corresponds to the direction of the external force, and the magnitude of the charge produced by the material directly correlates to the strength of the external force.^{75,76} Thus, all piezoelectric materials can convert mechanical energy into electrical energy through the piezoelectric effect. This allows them to gather vibration energy from various sources, such as the human body, which can help monitor human health.

Piezoresistivity refers to the change in the resistance of conductive materials when subjected to mechanical stress or strain.⁷⁷ The observed change in material resistance is primarily due to the structural deformation of the material and the piezoresistivity of the conductive materials themselves.^{78,79} Therefore, piezoresistivity is defined by the ratio of the fractional change in electrical resistance to the applied strain.

Alternatively, the triboelectric phenomenon can also be harnessed to detect small deformations. Figure 7B illustrates the working setup based on the triboelectric mechanism. In the initial condition (rest state), no charge is produced/induced. When two distinct materials (tribopositive and tribonegative) come into physical contact, triboelectric charges will be generated on the surfaces in contact. A potential difference can be created upon separating the two contacting surfaces, leading to an immediate flow of electrons from the bottom electrode to the top electrode due to the triboelectric effect and electrostatic induction. This electron flow continues until the two surfaces are entirely separated and reach a state of equilibrium. When the two materials are brought in contact again through external force, the electrostatically produced charges will return through the external load to balance the difference in electric potential. The electric current signal is produced from the entire working cycle.^{80,81}

These mechanisms can potentially measure HSs through the chest vibration caused by the HSs. Also, due to the simple fabrication process, ease of operation, flexibility, and cost-effectiveness, these techniques show promise for integration into wearable technology. It is worth noting, however, that their sensitivity may not reach the levels achieved by microphones or accelerometers.

3.3.2 | Sensing materials

Different materials are employed based on the sensing mechanism: piezoelectric, triboelectric and piezoresistive. Materials that can generate electrical charges in response to minor deformations induced by mechanical stimuli are termed piezoelectric materials. In contrast, piezoresistive materials exhibit variations in internal

resistance corresponding to deformation.^{82,83} By producing electric charges (i.e., an electric signal) autonomously, without reliance on an external power source, these transducers offer the advantageous feature of passive sensing.^{77,84} The selection of the appropriate sensing material is a pivotal consideration in the design of HSs sensors, governing the attainment of optimal sensitivity, frequency response, and stability.⁸⁵ Recent endeavors have explored various flexible materials, encompassing piezoelectric and piezoresistive variants, to propose HSs sensors that are lightweight, pliable, and conducive to long-term monitoring of cardiovascular function.

Piezoelectric materials can generate an electrical charge in response to applied mechanical stimuli. As quartz exemplifies, these materials, such as ceramics and polymers, can be artificially synthesized in a laboratory setting or occur naturally. In quartz's unit cell, the oxygen tetrahedron lattice is constructed around a central silicon atom, which imparts a distinctive atomic arrangement. Notably, the separations between the oxygen atoms and the distances between each oxygen atom and the silicon atom are uniform. However, when applying mechanical stress to the crystal, the atoms experience slight displacements from their equilibrium positions. This movement leads to a net dipole moment within the crystal, resulting in polarization and the generation of an electric field. The piezoelectric effect arises from the asymmetrical distribution of charged atoms within the crystal structure.⁸⁶

Artificial piezoelectric ceramics with crystal structures include lead titanate (PbTiO₃), barium titanate (BaTiO₃), lead zirconate titanate (PZT), lithium niobate (LiNbO₃), among others. However, their inherently brittle nature, costliness, and potential inclusion of hazardous substances limit their use in the design of flexible sensors. Despite these challenges, certain studies have been conducted. For instance, Chen et al.⁸⁷ devised a cantilever-structured HSs sensor utilizing an inorganic piezoelectric polymer, specifically lead zirconium titanate (PZT), achieving a high sensitivity of 9.2 V/g at frequencies below 1 kHz. Although patterned inorganic piezoelectric materials like lead zirconate titanate (PZT) or zinc oxide (ZnO) have been employed to design pliable piezoelectric devices, they still confront the above-mentioned constraints.⁷² Consequently, ongoing investigations have focused on piezoelectric materials possessing biocompatibility, cost-effectiveness, durability, and the capacity for facile integration into stretchable structures.

Polyvinylidene fluoride (PVDF), initially identified by Kawai,⁸⁸ is a commercially available polymer renowned for its piezoelectric and pyroelectric properties, mechanical resilience, and biocompatibility. Owing to

these attributes, PVDF finds extensive applications in sensors and batteries. With Young's modulus of 3.6 GPa, unpatterned PVDF lacks stretchability. Therefore, recent attention has turned towards patterned flexible PVDF-based electromechanical sensors. In HSs monitoring, PVDF with diverse structural configurations has gained prominence in constructing flexible sensors, primarily due to its cost-effectiveness, straightforward fabrication process, flexibility, and biocompatibility.⁷² Noteworthy contributions include the proposal of a flexible e-tattoo and a battery-free stretchable patch by Ha et al.⁷² (depicted in Figure 8A), as well as initiatives by Eyvazi Hesar et al.⁷³ Ji and Zhang⁸⁹ aimed to record HSs continuously utilizing PVDF film's serpentine structure. The resulting mechanoacoustic sensors exhibited remarkable stretchability, achieving sensitivities of 0.4 and 0.97 mV/ μe for thicknesses of 28 and 168 μm , respectively.

The internal structure of the material significantly influences the sensitivity of the sensor. To enhance sensor performance, flexible and wearable HSs sensors have been developed through modifications in material structure, such as fabrications involving porous fabrics and nanofiber arrangements. To augment sensor sensitivity and modulate the frequency response bandwidth of the material, Liu et al.⁹⁰ introduced an approach involving nanoscale and microscale pores in PVDF-TrFE, creating a bi-nano-micro porous dual ferro-electret hybrid self-powered flexible sensor (Figure 8B). This method harnessed the features of molecular β -phase dipoles in ferroelectric material and space charge dipoles to heighten sensitivity, attaining a piezoelectric coefficient (d_{33}) of 3312 pC/N, substantially surpassing that of typical piezoelectric materials.

Traditionally, fabrics were employed as acoustic absorbers due to their composition and structure. To create a sensitive audible microphone, Yan et al.⁹⁵ introduced a fabric synthesized from piezoelectric poly (vinylidene fluoride-trifluoroethylene) (P(VDF-TrFE)) loaded with barium titanate (BaTiO_3) ceramic particles, achieving a notably high piezoelectric charge coefficient (d_{31}) of 46 pC/N, thereby enhancing sensitivity. Similarly, due to its easy deformation upon small mechanical stimuli, the nanofiber structure significantly impacts the sensitivity of vibration/force/pressure sensors.⁹⁶ Consequently, Nayeem et al.⁹⁷ proposed a PVDF-based all-nanofiber structured ultrathin mechanoacoustic sensor for continuous HSs monitoring, demonstrating a high sensitivity of 10,050.6 mV/Pa in the frequency range of <500 Hz. Luo et al.⁹⁸ A sandwich-structured wearable and flexible nano HSs sensor was fabricated by synthesizing a P(VDF-TrFE)/ZnO/GR composite nanofiber film through the high-pressure electrospinning method (Figure 8C).⁹¹ The integration of ZnO and graphene-enhanced the β -phase

content of the composite, thereby improving overall sensing performance. Furthermore, Li et al.⁹² designed a highly stable and flexible HSs sensor utilizing near-field and far-field electrospinning and the triboelectric effect (Figure 8D). The proposed design achieved a 7027 mV/Pa sensitivity and could capture high-quality HSs signals at five different diagnostic auscultation points.

While PVDF and its copolymers are extensively explored for the fabrication of wearable HSs sensors due to their high piezoelectric coefficient, low cost, and flexibility, it is noteworthy that the pyroelectric properties of PVDF, specifically the generation of electric charges upon exposure to heat or high temperatures,^{99,100} may impact sensor sensitivity.¹⁰¹ Although the effect of these pyroelectric properties on HSs sensor sensitivity has not been thoroughly examined, it holds conceptual relevance.¹⁰² In light of this, Poly(L-lactic acid) (PLLA), categorized as a piezoelectric polymer,¹⁰³ has emerged as an alternative for stethoscope fabrication. The PLLA-based sandwich-structured film stethoscope (PFS) achieved a sensitivity of -53.4 dB at 1 kHz, with a stable frequency response within the 100 Hz to 4 kHz range. Given its lack of pyroelectric properties compared to PVDF, PLLA is a viable option for selecting an appropriate sensing material for highly sensitive HSs sensors.¹⁰¹

In addition to the piezoelectric mechanism, the potential of piezoelectret polymers, such as fluoroethylene propylene (FEP), for detecting acoustic signals emitted by the human body has also been investigated. These polymers demonstrate macroscopic piezoelectricity, which arises from the displacement of stored charges and the phenomenon of electrostatic induction.¹⁰⁴ Han et al.¹⁰⁵ developed an FEP-based folded double-layer structure of flexible sensors to measure heart, Korotkoff, and breath sounds. The designed sensor demonstrates a notable sensitivity of 591 pC/kPa within the 0–8 kPa range. Additionally, it possesses a broad frequency bandwidth ranging from 0 to 600 Hz while maintaining exceptional stability throughout 1.1 million cycles.

Piezoresistive materials exhibit a change in electrical resistance in response to mechanical strain or stress. While much attention has been directed toward piezoelectric materials for HSs detection, less emphasis has been placed on applying piezoresistive materials. However, piezoresistive sensors offer several benefits, including a simple design and easy fabrication process at a low cost.¹⁰⁶ Graphene, a crystalline carbon allotrope comprising a single layer sheet of sp^2 hybridized carbon atoms, has garnered significant interest due to its exceptional electrical, mechanical, chemical, magnetic, and optical properties.^{107,108} To enhance its sensitivity while preserving mechanical stability and mobility, graphene is often engineered into three-dimensional (3D) porous structures,

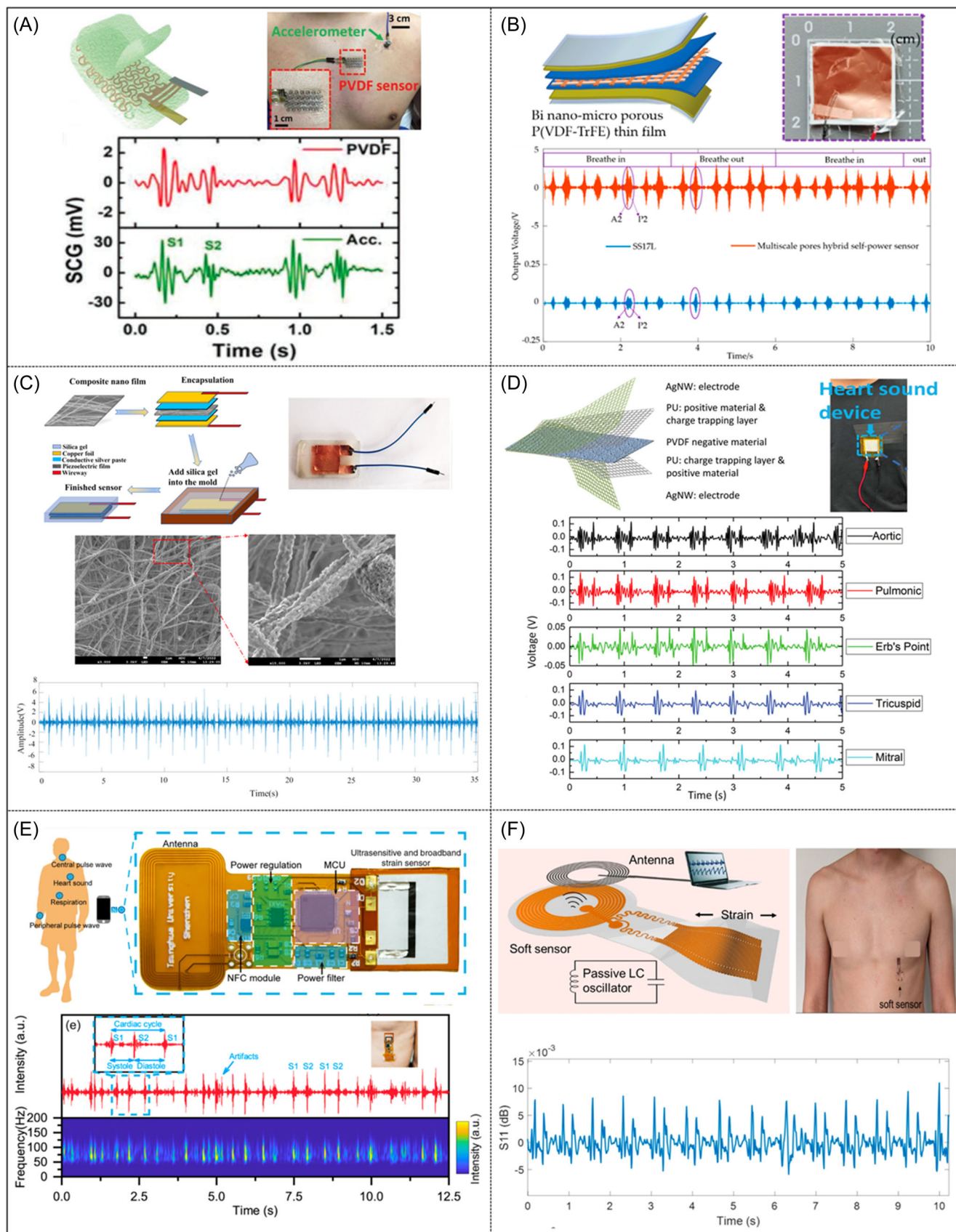


FIGURE 8 (See caption on next page).

augmenting its surface area.^{109,110} Conventional methods for generating 3D graphene structures involve creating foam from graphene oxide (GO), though this process is hindered by the need for GO precursor production via oxidative acid synthesis.^{111–113} An alternative approach involves chemical vapor deposition (CVD) of graphene onto a porous substrate to form the 3D structure. Yet, high-temperature conditions and subsequent etching and drying procedures may impede scalability.^{114–116} In 2014, Lin and colleagues¹¹⁷ introduced a breakthrough technique, employing a CO₂ infrared laser to write on polyimide directly (PI) film, creating laser-induced graphene (LIG). This method, characterized by its simplicity, speed, cost-effectiveness, and capacity for customizable patterning using computer-based design, has been explored to develop HSs sensors. Liu et al.⁹³ proposed a wireless, battery-free wearable patch that monitors various physiological signals within the 0.2–200 Hz frequency range. Through adjustments to the thickness and microstructures of the LIG/polydimethylsiloxane film, they achieved a maximum gauge factor of 456.51. A flexible circuit utilizing near-field communication (NFC) technology facilitated the wireless power transmission, as illustrated in Figure 8E. Similarly, Yang et al.¹¹⁸ investigated a wearable strain sensor based on lignin-derived LIG, demonstrating high sensitivity to minute deformations caused by vibrations and small-scale human body movements. The sensor successfully detected HSs.

In addition to the previously discussed piezoelectric and piezoresistive flexible sensors, Guess et al.⁹⁴ introduced a wireless, passive, and battery-free soft capacitive sensor that measured HSs using inductive coupling (Figure 8F). The sensor and coil were seamlessly integrated onto a single layer of copper, supported by PI, streamlining the design process, and eliminating complexity. A low modulus and biocompatible silicone gel for encapsulation ensured safe attachment to human skin with no adverse effects. The passive nature of the sensor, devoid of integrated batteries and IC components, conferred high flexibility and stretchability. The designed sensor was compared with a commercial accelerometer and single-lead ECG to validate its functionality. The results exhibited a

robust correlation when measuring fiducial points such as mitral valve closing (MC), mitral valve opening (MO), isovolumic movement (IM), aortic valve opening (AO), and aortic valve closing (AC). A summary of all proposed wearable and flexible HSs sensors is presented in Table 3.

In conclusion, the array of piezoelectric, piezoresistive, piezoelectric, triboelectric, and capacitive flexible and wearable sensors explored above hold tremendous promise for applications in home health-care, remote health monitoring, and self-diagnosis. Their lightweight design, ease of use, suitability for wearables, and cost-effectiveness are notable advantages. However, efforts to enhance their operational range, sensitivity, stability, durability, and biocompatibility are imperative for enabling accurate, long-term monitoring. Exploring novel materials like flexible polymers, metal-organic framework (MOF), carbon-based biomaterials¹¹⁹ and their composites is crucial for expanding the capabilities of health sensing devices. Integrating composite materials and harnessing the strengths of multiple components is pivotal for advancing health sensing technologies. Composites such as PVDF/ZnO, PVDF/BaTiO₃, and PVDF-TrFE/ZnO/GR have demonstrated enhanced sensitivity, underscoring the potential of composite materials in sensor improvement. Additionally, research into environmentally friendly, green biomaterials has the potential to improve the biocompatibility of wearable sensing technologies while reducing the energy consumption and environmental impact associated with material manufacturing and disposal. Further innovations in composite materials, including unique structural configurations, surface functionalization, and biocompatibility, will propel health sensing technologies' evolution.

4 | HSS FILTERING TECHNIQUES

The HSs signal refers to a physiological signal that is relatively weak compared to the significant ambient noise. The HSs signal is very sensitive to various human

FIGURE 8 Flexible polymers-based HSs sensors. (A) Serpentine structured PVDF-based piezoelectric wearable sensors for HSs (Reproduced under the terms of CC-BY 4 license: Copyright 2019, Wiley-VCH.⁷²). (B) Bi nano-micro porous structure-based HSs sensor (Reproduced under the terms of CC-BY 4 license: Copyright 2021, MDPI.⁹⁰). (C) P(VDF-TrFE)/ZnO/GR composite nanofiber film based flexible nano HSs sensor (Reproduced under the terms of Beilstein-Institut Open Access License.⁹¹). (D) Triboelectric effect-based grid structured device for HSs detection (Reproduced under the terms of CC-BY-ND-NC license: Copyright 2023, Wiley-VCH.⁹²). (E) LIG-based wearable patch for low-frequency physiological signal monitoring (Reproduced with permission: Copyright 2021, ACS Publications.⁹³). (F) Wireless, and battery-free soft capacitive HSs sensor (Reproduced under the terms of CC-BY 4 license: Copyright 2023, OAE.⁹⁴). HS, heart sound; LIG, laser-induced graphene; PVDF, polyvinylidene fluoride.

TABLE 3 Summary of polymers-based flexible and wearable HSS devices.

Material	Structure	Mechanism	Sensitivity	Application	Reference
PVDF	Solid film	Piezoelectric	0.40 mV/ $\mu\epsilon$	HS and cardiac time intervals	[72]
PVDF	Solid film	Piezoelectric	–	Continuous recording of HSs	[73]
PZT	Cantilever	Piezoelectric	9.20 V/g	Heart and lung states of pneumonia patients	[87]
PVDF	Solid film	Piezoelectric	0.97 mV/ $\mu\epsilon$	Monitoring of respiratory and heartbeat	[89]
BaTiO ₃ -PVDF-TrFE	Fabric	Piezoelectric	–	HSs	[95]
PVDF	Nanofiber	Piezoelectric	10050.60 mV/Pa	Continuous long-term heart monitoring	[97]
PVDF-TrFE	Porous	Piezoelectric	–	HSs	[90]
PVDF-TrFE/ZnO/GR	Nanofilm	Piezoelectric	–	HSs	[91]
PVDF/PU	Nanofiber	Triboelectric	7027 mV/Pa	HSs	[92]
PLLA	Solid film	Piezoelectric	–	Heart and pulse signal	[101]
FEP	Groove surface	Piezoelectric	591 pC/kPa	Heart, breath, and Korotkoff sounds	[105]
LIG	Microstructure	Piezoresistive	GF- 456.51	Multiple physiological signal measurement	[93]
LIG	Porous	Piezoresistive	GF- 960	HSs	[118]
Copper	Soft film	Capacitive	2 MHz per 1% ϵ	HSs	[94]

Abbreviations: HS, heart sound; PVDF, polyvinylidene fluoride.

factors due to their intrinsic instability as natural signals, which are generated by the intricate processes of life. The initial symptoms of abnormalities in the heart include the appearance of cardiac murmurs and changes in HSs. Therefore, examining the accurate HSs is crucial for understanding the condition of the heart and blood arteries.

Noise in the HSs has been identified as a valuable indicator for identifying VHD or CVDs early. The presence of noise at a different stage of HSs signal identifies various types of cardiac diseases, as shown in Figure 9A–D. Thus, it is necessary to record HSs in a noiseless environment to monitor and diagnose cardiac diseases accurately. However, the frequency range of HSs generated by the opening and closing of atrioventricular and semilunar valves is notably low, spanning from 20 to 250 Hz, rendering it highly susceptible to interference from noise. Therefore, to accurately diagnose CVDs by recording a noise-free HSs pattern, employing a sensor with ultrahigh sensitivity in conjunction with intelligent and highly efficient signal processing tools is imperative. Initially, the signal captured by the sensor, constituting the HSs signal, is considered raw due to the significant noise influence. Typically, primary amplification is conducted to enhance the HSs signal alongside the noise, followed by applying low pass, high pass, or bandpass filters to isolate the noise. The filtered signal then undergoes secondary amplification for further analysis. Various amplifiers can be utilized for this purpose, and a range of filtering techniques and algorithms have been

explored to separate noise from the original signal effectively.

Digital bandpass filtering techniques are extensively employed in HSs devices, allowing only specific frequency ranges to pass through while attenuating frequencies outside of this range.¹²¹ The widespread use of digital bandpass filters stems from their distinct advantages: precise control, reproducibility, adaptability, and stability. These qualities render them indispensable tools in signal processing applications spanning audio, image, and biomedical signal processing.¹²² Consequently, diverse digital bandpass filters have been applied in the literature for processing HSs signals, particularly for denoising and filtering purposes. Among these, the 4th/3rd order Butterworth bandpass filter is the most prevalent choice.^{25,26,34,40,48,62,68,72,93} In addition to this, alternative methods such as the MATLAB plugin FDATool,⁹⁰ notch filter,⁹⁷ normalized least mean square adaptive filter,¹²³ Chebyshev filter,^{52,124} Hilbert FIR filter,⁴⁷ hardware filter bank, and wavelet denoising algorithm,^{34,40,65,125} 3rd order Savitzki Golay Filter,³⁶ empirical mode decomposition (EMD),⁴³ Bessel filter,⁶ and Kaiser window filter^{126–128} have also been explored for HSs signal processing.

Besides digital filtering, extensive research has explored various techniques including statistical, time domain, and frequency domain methods for extracting noise from HSs. Some of the most potent methods include nonnegative matrix factorization (NMF) with adaptive contour representation computation using the

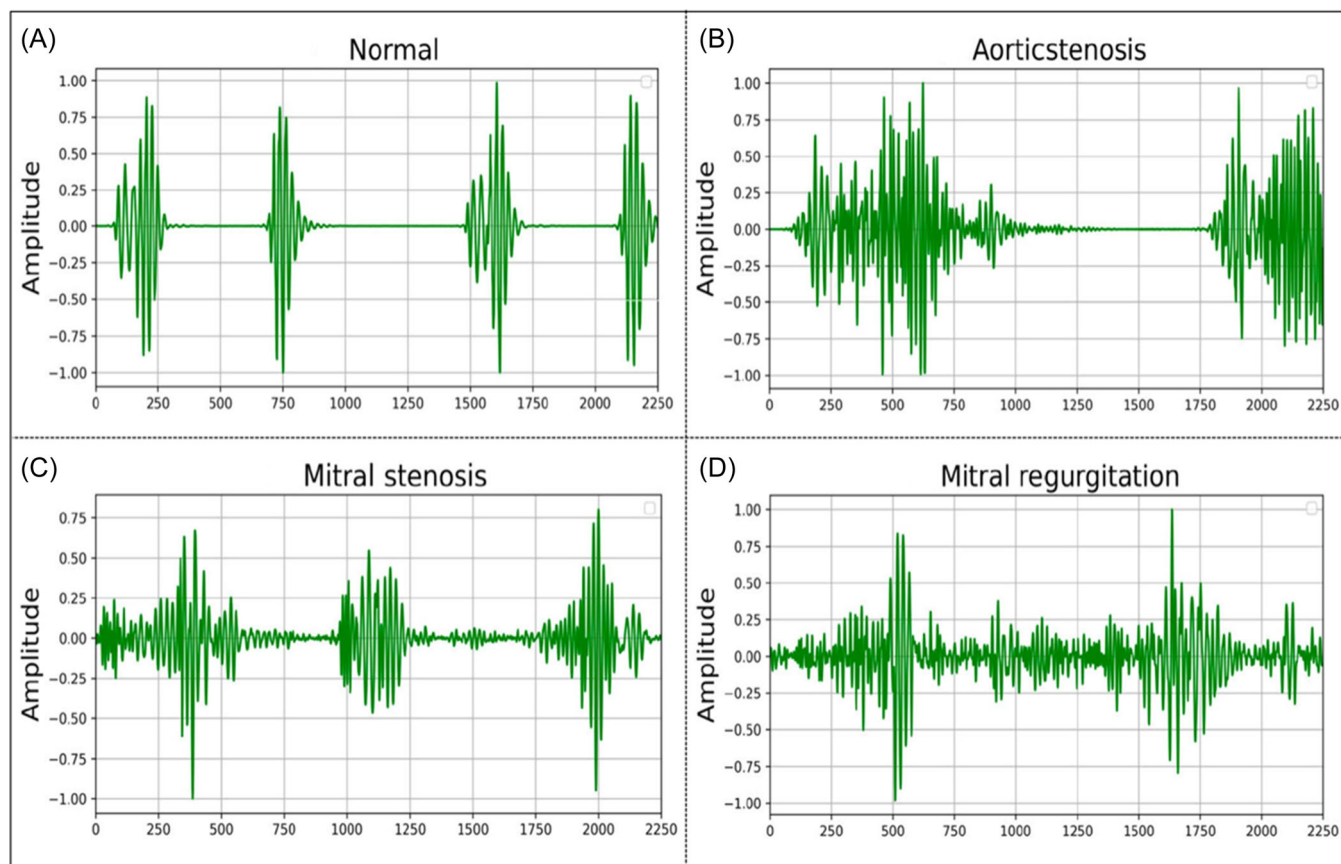


FIGURE 9 Impact of noise in HSs signal for the diagnosis of diseases. (A) Normal HSs signal. (B) Aortic stenosis. (C) Mitral stenosis. (D) Mitral regurgitation (Reproduced under the terms of CC-BY 4 license: Copyright 2021, IEEE.¹²⁰). HS, heart sound.

corresponding spectrogram,¹²⁹ EMD,¹³⁰ and the integration of singular value decomposition and compressed sensing.¹³¹ It is vital to evaluate actual denoising algorithms since they may introduce anomalies that decrease classification precision. A novel deep-learning framework was proposed for the real-time denoising of noisy HSs recordings.¹³² This framework can remove external noises from the recordings, such as lung sounds and hospital ambient noise. The suggested framework maintains the structure of the signals, particularly murmurs, which are crucial markers of heart problems, even when exposed to severe input distortions.

Furthermore, a novel method for retrieving cardiac auscultation sounds was introduced, utilizing a modified cosine transform followed by a long short-term memory (LSTM) based masking technique.¹³³ The suggested methodology presents a promising approach to effectively separate HSs and murmurs from ambient noise and other disruptive sounds. Alternatively, other strategies such as single-channel blind source separation,¹³⁴ U-Net¹³⁵ and multi-channel U-Net (MCUNet)¹³⁶ have also been explored for the denoising of HSs signal.

HSs analysis and processing are vital for identifying important features and improving clinical evaluations. To extract accurate information from the noisy HSs signal, several filtering methods have been employed. Every filtering method has pros and cons; choosing the best depends on the application. Time-domain filtering methods like Chebyshev, Bessel, and Butterworth are simple and computationally efficient, making them ideal for real-time applications. Frequency-domain filtering methods like the Fourier transform have eliminated noise-related frequency components. Wavelet-based filtering improves temporal and spectrum localization, preserving HSs signal transitory properties.

In addition to HSs, lung sound denoising is crucial for comprehensive cardiovascular monitoring. Lung sounds, including normal breathing sounds, wheezes, and crackles, provide valuable information about respiratory health often linked to CVDs. Specific alterations in breath sounds may indicate conditions such as pulmonary edema, usually related to heart failure.¹³⁷ However, accurate diagnosis is hampered by ambient noise and patient movements, requiring effective denoising and analysis of these sounds.

Advances in sensor technology and signal processing techniques, proven effective in denoising HSs, can be leveraged to improve the clarity and reliability of lung sound analysis. Techniques such as EMD, wavelet decomposition, and machine learning algorithms have shown promise in processing breath sounds. For example, EMD and wavelet decomposition have been used in breathing sound processing to extract relevant information from lung sounds.^{138,139} Additionally, machine learning algorithms have been used for respiratory sound detection, thereby facilitating the diagnosis of respiratory diseases.^{140,141} Additionally, research has focused on methods such as NMF clustering and deep learning algorithms to separate heart and lung sounds, contributing to more accurate analysis.¹³⁴ These approaches differentiate heart and lung sounds based on their distinct characteristics, thereby improving diagnostic capabilities. Furthermore, studies have explored applying various signal processing techniques, such as multi-scale entropy and fractal dimension analysis, to classify lung sounds and detect abnormalities.^{142,143}

In conclusion, integrating advanced sensor technologies and signal processing methods into analyzing heart and lung sounds is essential for a comprehensive approach to cardiovascular monitoring. By adapting HSs denoising techniques and incorporating innovative methods such as EMD, wavelet decomposition, and machine learning algorithms, healthcare professionals can improve the accuracy and efficiency of diagnosing conditions. respiratory diseases, which often coexist with CVDs.

5 | APPLICATIONS OF HSS

HSs encompass valuable information regarding cardiac activity and conditions, facilitating the early detection of CVDs. Consequently, HSs are a diagnostic tool for ailments impacting blood vessels, heart valves, and circulation. Furthermore, they offer a means of alleviating distress for affected individuals, potentially aiding in preventing heart attacks. HSs analysis contributes to detecting and assessing heartbeat patterns and variability in heart rate.^{144–147} Additionally, it can be integrated with other physiological signals like ECG and PPG to estimate hemodynamic parameters, including systolic and diastolic blood pressure (DBP), stroke volume (SV), and cardiac output (CO).^{148–152} HSs play a crucial role in diagnosing conditions such as coronary artery disease and atrial fibrillation¹⁵³ and distinguishing between inhalation and exhalation phases.¹⁵⁴ This section will delve into studies employing HSs signals for clinical applications.

5.1 | Diagnosis of CVDs

Sleep apnea denotes recurrent episodes of obstructive apnea and hypopnea during sleep, often accompanied by daytime sleepiness or compromised cardiopulmonary function.¹⁵⁵ With over 900 million individuals worldwide affected by obstructive sleep apnea, its untreated progression can lead to significant health complications, yet 80% of cases remain undiagnosed.¹⁵⁶ Zavanelli et al.²⁶ devised a pliable, wireless, fully integrated patch with skin-like properties to address this. Positioned at the sternum, this device adeptly captures HSs, ECG, and PPG signals. In initial trials involving symptomatic subjects and controls, the apparatus demonstrated remarkable proficiency in discerning diverse parameters, including blood-oxygen saturation, aortic opening mechanics, heart rate, respiration rate, cardiac pre-ejection period (PEP), ejection timing, and heart rate variability. In preliminary at-home assessments with symptomatic patients, machine learning algorithms were leveraged to identify apnea and hypopneas autonomously, achieving an impressive sensitivity of 100% and an accuracy of 95%.

Transesophageal Doppler (TED) employs the Doppler methodology to assess aortic blood flow velocity as it descends. This velocity-time curve is the basis for deriving crucial hemodynamic parameters, including CO, preload, afterload, and myocardial contractility.¹⁵⁷ While TED is a validated technique, it offers minimal invasiveness for assessing SV. In light of this limitation, Semiz et al.¹²⁷ introduced a wearable device affixed to the mid-sternum of postsurgical patients, enabling continuous acquisition of HSs and ECG signals to estimate SV. Furthermore, various machine learning approaches have been developed employing raw HSs data to surveillance and classify CVDs, distinguishing between normal and abnormal sounds, aortic and mitral stenosis, and mitral regurgitation.^{6,120,158}

Cardiac murmurs are crucial in diagnosing pathological cardiac conditions, including arrhythmia, heart failure, and valve disease. For instance, Qu et al.⁶⁸ discerned distinct patterns of cardiac murmurs throughout the cardiac cycle, enabling the diagnosis of different heart diseases such as mitral stenosis (narrowing of the mitral valve), patent ductus arteriosus (unclosed hole in the aorta artery), mitral regurgitation (blood leakage backwards through the mitral valve), aortic insufficiency (aortic valve dysfunction), and aortic stenosis (narrowing of the aortic valve). Likewise, Liu et al.²⁵ investigated diverse HS patterns emanating from the tricuspid and pulmonary valve positions in patients diagnosed with mild tricuspid and pulmonary regurgitation. Thus, segmenting HSs and murmurs is paramount in developing an HSs classification system for precise heart state

predictions. This, in turn, assists clinicians in assessing the patient's cardiac condition and determining timely therapeutic interventions.

Besides CVDs, Pneumonia, an inflammatory ailment stemming from a lung infection, typically undergoes resolution within 2–4 weeks.¹⁵⁹ However, infants, older people, and individuals with pre-existing heart or lung conditions necessitate vigilant monitoring and specialized care.¹⁶⁰ Consequently, assessing cardiac and pulmonary functions is paramount for pneumonia patients. Chen et al.⁸⁷ introduced a compact, highly sensitive accelerometer-based apparatus to continuously monitor heart and lung conditions. This innovation holds the potential to discern cardiac and pulmonary distress, further enabling the tracking of recovery progress in pneumonia patients after their discharge.

5.2 | BP estimation

HSs signals contain crucial physiological events (feature points) that facilitate the estimation of hemodynamic variables, with BP being a pivotal parameter for monitoring CVD patients. While intra-arterial measurement is the most precise method for BP determination, its invasive nature restricts widespread application.^{161,162} Various noninvasive, cuff-based, and cuffless technologies have emerged for BP measurement to overcome this limitation. These include sphygmomanometers,¹⁶³ tonometry,^{164–166} PPG, and ECG methods.^{167–169} Additionally, researchers have explored specific HSs features for noninvasive, cuffless BP estimation using machine/deep learning approaches. Table 4 provides an overview of fundamental HSs features documented in the literature with a graphical representation shown in Figure 10A. These features of HSs have great potential for the diagnosis of CVDs. This section will delve into recent studies concerning BP estimation utilizing HSs.

Pulse transit time (PTT), signifying a pressure wave's duration to traverse two arterial points, has been extensively investigated for BP estimation employing PPG/ECG signals.^{178–181} Moreover, diverse PTT measurement methodologies have been introduced concerning HSs signals. For instance,¹⁸² introduced a novel PTT measurement approach utilizing a logarithmic model. This method employs the AO peak within the HSs signal as a proximal indicator and finger and ear PPG signals as distal indicators. Similarly,¹⁸³ developed a model utilizing HSs, PPG, and ECG signals, capable of gauging PTT as the temporal interval between AO and the initiation of the PPG upslope recorded at the finger site.

The computation of PTT from either a single HSs signal or a combination of HSs with PPG/ECG signals

has been explored for BP estimation. For instance,^{184,185} and¹⁸⁶ employed a smartphone's accelerometer, a microphone connected to the diaphragm, an accelerometer for HSs recording, and a phone camera alongside an optical sensor for PPG signal recording, respectively, to derive PTT for noninvasive cuffless BP estimation. In,¹⁸⁵ BP was estimated by calculating PTT by taking the S1 peak of HSs as the proximal timing reference and different characteristic points of the PPG signal (i.e., start to rise PTTf, maximum slope PTTd and peak PTTp) as a distal timing reference. Estimation with PTTf and PTTd was inaccurate, so only the PTTp characteristic point was used for the BP estimation. It achieved 0.84 and 0.86 as the correlation coefficient (r) of SBP and DBP estimation, respectively.

Similarly,¹⁸⁷ devised a wearable watch-based, non-invasive cuffless BP device that estimates BP using PTT measured from HSs, ECG, and PPG signals. PTT was computed as the difference between the proximal timing reference (i.e., AO) and the distal timing reference (i.e., foot of the highest PPG peak). This method achieved the root mean square error (RMSE) of 2.72 mmHg and 0.69 Pearson correlation coefficient. A study by Lee et al.¹⁵⁰ investigated a novel BP and pulse pressure (PP) estimation method based on PTT and SV approximation using the PEP calculated from HSs, ECG, and PPG signals. The PEP interval was obtained by measuring the time between the ECG's R-peak and the SCG's AO-peak.

Meanwhile, the PTT was calculated by measuring the time between the AO peak of the SCG and the minimum

TABLE 4 Basic features of human HSs signal.

Features	Abbreviations	Reference
MO	Mitral valve opening	[170–172]
MC	Mitral valve closure	[170–172]
AO	Aortic valve opening	[170–172]
AC	Aortic valve closure	[170–172]
IC	Isovolumic contraction	[170]
IM	Isovolumic movement	[170]
IVRT	Isovolumic relaxation time	[172, 173]
IVCT	Isovolumic contraction time	[172, 173]
LVET	Left ventricular ejection time	[173–175]
PEP	Pre-ejection period	[173–175]
AS	Peak of atrial systole	[170]
RE	Onset of rapid ejection	[170–172]
RF	Peak of rapid ventricular filling	[170, 171]
STI	Systolic time interval	[173, 174]

Abbreviation: HS, heart sound.

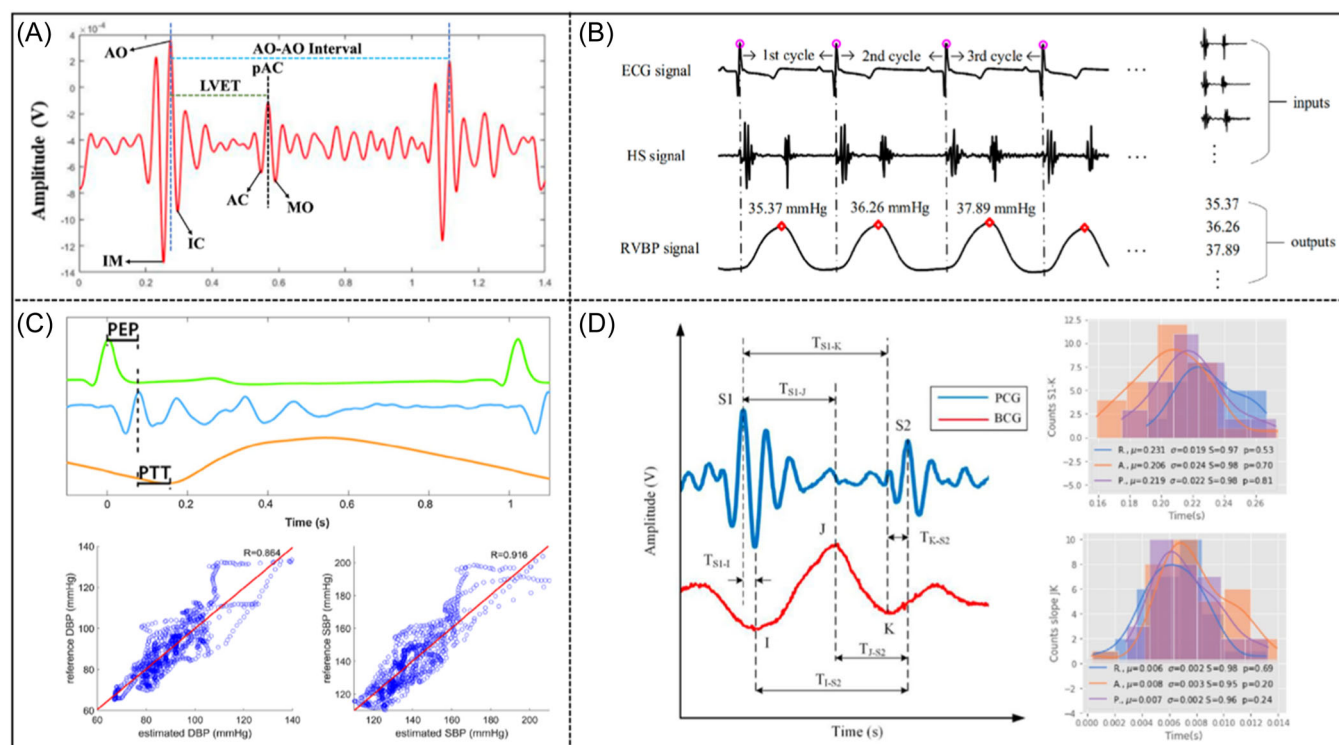


FIGURE 10 Different methodologies for the estimation of BP using HSs. (A) Essential characteristics points of HSs signal (Reproduced under the terms of CC-BY 4 license: Copyright 2022, MDPI.³³). (B) RVBP estimation using ECG and HSs (Reproduced under the terms of CC-BY 4 license: Copyright 2020, MDPI.¹⁷⁶). (C) BP and PP estimation based on PTT and SV approximation using PEP from ECG, HSs and PPG signals (Reproduced under the terms of CC-BY 4 license: Copyright 2018, Springer Link.¹⁵⁰). (D) Estimation of SBP from the time interval between S1 and S2 peaks of HSs signal and I, J and K points of BCG (Reproduced under the terms of CC-BY 4 license: Copyright 2022, Springer Nature.¹⁷⁷). ECG, electrocardiogram; HS, heart sound; PTT, pulse transit time.

point of the PPG waveform (Figure 10C). This approach employed the 2-element Windkessel model to characterize BP and PP, relating it to the ratio of SV and arterial compliance (C). The proposed model attained high accuracy for estimating PP, DBP, and SBP, i.e., RMSE 3.46, 6.31, and 8.41 mmHg, respectively. In addition to PTT-based BP estimation,¹⁸⁸ presented a model for BP estimation based on the PEP using HSs signal in conjunction with ECG and PPG. PEP was computed as the time delay between the ECG-R peak and the AO peak of the HSs. In contrast, pulse arrival time (PAT) was estimated as the time delay between the R-peak of ECG and the rise point of PPG. Finally, PTT was measured to calculate BP by subtracting PEP from PAT.

Since approximately 86.29% of the global population are smartphone users^{189,190} introduced novel smartphone applications for BP measurement. These applications collected HSs and finger pulses using phones and an external microphone. BP estimation was performed by computing the vascular transit time (VTT) based on the recorded HSs and finger pulses with 95%–100% accuracy, utilizing two phones and one phone equipped with an external microphone. In parallel,¹⁹¹ and¹⁹² explored the

utility of the S2 for noninvasive, cuffless, and continuous BP estimation. They employed multivariate regression analysis and a support vector machine approach to process the HSs collected through a diaphragm-equipped stethoscope. The mean absolute error (MAE) and the mean error (ME) of multivariate regression analysis were 6.08 and 0.31 mmHg, respectively. On the other hand, ME and standard deviations (SDs) for the support vector machine approach were less than 5 and 8 mmHg, respectively.

Furthermore, Gonzalez-Landaeta et al.¹⁷⁷ investigated the BCG's I, J, and K parameters in conjunction with S1 and S2 peaks of HSs. Both S1 and S2 peaks serve as proximal time references, allowing for estimating time delays with each I, J, and K wave of the BCG, as presented in Figure 10D. Furthermore, the periods mentioned above are explanatory variables within a Random Forest regressor (RF) to estimate SBP. The study revealed a negative association between increased SBP and intervals S1-J, S1-K, S1-I, J-S2, and I-S2. Among the different regression algorithms assessed for BP estimation, including simple linear regression, support vector machine, kernel ridge, and Laplacian kernel function,

Random Forest achieved a mean error of 3.3 mmHg, with a SD of ± 5 mmHg. This approach holds promise for integration into wearable devices employing cost-effective embedded technologies.

Regular examination of right ventricular systolic blood pressure (RVSBP) proves crucial for early detection of pulmonary hypertension (PH).^{193,194} In their study, Wang et al.¹⁷⁶ propose a deep learning model designed for the noninvasive estimation of RVSBP, utilizing both HSs and ECG signals, employing five beagle dogs as subjects (Figure 10B). To induce a wide range of RVSBP variations, U-44069 medicine was administered. The relationship between RVSBP and HS signals was effectively modeled using the bidirectional LSTM network, resulting in a MAE of 6.37 ± 4.90 mmHg, considering the SD.

The above-discussed models for noninvasive BP estimation typically necessitate using more than one physiological signal, such as PPG/ECG, HSs/PPG, HSs/ECG, or ECG/PPG/HSs, which can introduce complexity into the system. To address this, Das et al.¹⁴⁸ introduced an LVET-based BP estimation model that exclusively relies on HSs signals. Like the PTT, the LVET is a function of pressure; therefore, pressure parameters can be inferred from it. The LVET gives systolic time intervals between aortic flow ejection and termination. Blood leaves the heart at the AO moment, making it a pressure pulse proximal point. At the AC instant, blood reaches the measurement sites and returns from the body to the heart, ending the pulse. This support using LVET instead of PTT to determine BP. The significant structural heterogeneity of HSs among participants makes AC identification challenging, and body movements, respiration, and other external noises can distort the HSs signal. The suggested model calculates the LVET interval using pAC peak time (peak following the AC point). They utilized a MEMS-based accelerometer to record HSs signals. Subsequently, they estimated both systolic blood pressure (SBP) and DBP by calibrating them with the subjects' original BP values. The proposed method demonstrated mean errors for SBP and DBP within the range of -0.197 ± 3.332 and -1.299 ± 2.578 mmHg, respectively. These calculated BP estimation errors conform to the conditions outlined by the IEEE standard, specifically within 5 ± 8 mmHg. Therefore, this method holds promise for enabling ubiquitous, continuous BP monitoring.

Various artificial intelligence techniques have been proposed for noninvasive cuffless BP estimation. However, these methods face challenges in accurately predicting BP for subjects in motion or during exercise. Addressing this issue of motion artifact, Yang et al.¹²³ conducted a study on the impact of dual MEMS accelerometers on motion artifact cancellation. Their findings

revealed that their HSs readings exhibit slight variations when two accelerometers are strategically positioned on the chest. At the same time, the motion artifacts produced are nearly identical. By subtracting the data from both accelerometers, motion artifacts are effectively eliminated, enabling the emergence of distinct HS features.

Similarly, Hsu et al.¹⁹⁵ applied a comparable approach to developing a model resilient to motion artifacts for BP estimation. They employed a sequence of accelerometers placed on the subjects' heart, sternum, bilateral wrists, and left carotid. This technique successfully removes motion artifacts and extracts features from the multidimensional HSs. In the seated position, the mean and SD of systolic and diastolic difference errors (DE) were found to be -0.25 ± 5.45 and 0.09 ± 4.10 mmHg, respectively. Furthermore, the mean absolute difference errors (MADE) were recorded as 4.73 and 3.62 mmHg. During motion (walking), the DE values were -0.38 ± 6.67 for SBP and 1.15 ± 4.47 mmHg for DBP, with corresponding MADE values of 5.07 and 3.36 mmHg. Notably, the model complies with the British Hypertension Society protocol and AAMI standard, demonstrating the practicality of portable BP monitoring within healthcare facilities.

HSs analysis offers considerable promise as a non-invasive method for the early detection of CVDs. The processing of HSs provides valuable observations regarding cardiac function and enables the measurement of essential hemodynamic parameters. Thus, wearable HSs devices offer advantages over conventional CVD measurement techniques due to their noninvasive nature, cost-effectiveness, and convenient accessibility. Furthermore, these devices have the potential of continuous BP monitoring, which is very useful in situations calling for frequent measurements or in people whose BP levels fluctuate often. Despite the proposed models, the investigation of HSs to estimate BP remains an area that is currently in the process of development. Validating and developing existing approaches, refining algorithms, and overcoming challenges like noise interference and individual variability, results validation, and system calibration can be improved in the future. The power of this technology to offer continuous, accurate, and convenient measurements has the potential to fundamentally revolutionize the field of remote health monitoring.

6 | DISCUSSION

Besides ECG and BP measurements, which stand as primary and reliable diagnostic tools for CVDs, it is worth noting that HSs generated by the cardiac valve

movements during the cardiac cycle offer a potential avenue for early CVD detection. Although stethoscopes are conventionally employed for HS monitoring, they are constrained because only an experienced clinician can interpret the auscultated sounds for diagnostic purposes. However, with the advancement in wearable HSs devices, it has become possible to diagnose CVDs early. Such devices serve as diagnostic markers for CVDs and can perform long-term remote monitoring and identify and document subtle changes in cardiac function.

Recent advancements in health sensing (HSs) devices have been remarkable in response to evolving application demands. These sensors have evolved from rigid counterparts like accelerometers, microphones, sound sensors, and force sensors to cutting-edge thin polymer-based models, offering flexibility and wearability, as outlined in Tables 1–3. While rigid or partially soft devices, such as IC chip or MEMS-based sensors, boast high sensitivity, accuracy, stability, and durability, they suffer from drawbacks like patient discomfort during prolonged use, skin irritation, and limited conformity, hampering their wider deployment. This progression signifies a substantial leap in complexity, effectively addressing the diverse needs of numerous applications. By leveraging biocompatible materials, flexible devices can mitigate issues related to comfort, mobility, skin irritation, and conformity limitations. Over recent years, there have been significant strides in novel materials, sensing mechanisms, and structural designs for flexible and wearable HSs sensors, resulting in vastly improved sensing performance. Nonetheless, challenges persist regarding stability and durability, particularly for continuous long-term monitoring with flexible devices.

Green biomaterials are derived from natural sources and are designed to minimize the environmental impact throughout their lifecycle, from production to disposal. The vitality of such materials in modern technologies has grown due to concerns about environmental issues and the need for sustainable practices across industries, including healthcare. Green biomaterials have emerged as a promising alternative in wearable medical devices due to their biocompatibility, biodegradability, and reduced carbon footprint.¹⁹⁶ Natural biomaterials such as silk fibroin, spider silk, cellulose, and chitosan are examples of biomaterials that meet the principles of green biomaterials.¹⁹⁷ The use of sustainably produced biomaterials can enhance the biocompatibility of wearable sensors while reducing the energy consumption and environmental impact associated with material manufacturing and disposal.¹⁹⁸ These biomaterials are abundant, sustainable, biocompatible, and biodegradable, making them ideal for applications in wearable pressure sensors and other wearable electronic devices.¹⁹⁹

Furthermore, the development of natural biomaterials holds great promise for advancing the development of high-performance, sustainable epidermal electronics, highlighting the importance of transitioning to eco-friendly materials in this field.²⁰⁰

In the context of wearable technology, the integration of green biomaterials can drive technological advancements in various fields, demonstrating the potential of these materials to revolutionize the design and functionality of wearable devices. The shift toward utilizing natural piezoelectric biomaterials, which are known for their biocompatibility, biosafety, and environmental sustainability, highlights the shift from traditional materials to more sustainable options in emerging fields such as biomedical devices.²⁰¹ Furthermore, the development of degradable piezoelectric biomaterials for wearable and implantable bioelectronics marks a paradigm shift toward utilizing green, degradable functional materials with enhanced biocompatibility.²⁰²

The advancement of next-generation wearable and flexible HSs sensors is pivotal for continuous and real-time monitoring of heart activity, revolutionizing digitized medical practices. Specifically, acoustic sensors like microphones or accelerometers utilized for digitally capturing human body sounds have the potential for designing more compact and intricate ICs. Integrating the pre-amplification stage into the chip's sensing circuitry can significantly reduce device size and required components. Moreover, the significant progress in soft and flexible electronics has facilitated the development of innovative wearable HSs devices boasting superior mechanical properties, smaller form factor, and enhanced compatibility with the skin compared to traditional rigid devices.

In the multifaceted realm of HSs sensors, meticulous consideration of various parameters is indispensable to ensure their efficacy and efficiency across diverse applications. Factors such as long-term stability in challenging environmental conditions like water, heat, or salt, power consumption, cost-effectiveness, and feasibility of mass production warrant thorough examination. Environmental stability is paramount for HSs sensors, given their exposure to diverse conditions; thus, employing resilient materials and effective encapsulation designs is essential to ensure sustained functionality over extended periods. Additionally, minimizing power consumption is crucial, especially considering the integration of HSs sensors into Internet of Things (IoT) networks. However, achieving low power consumption remains complex, necessitating meticulous sensor and system design refinement across multiple domains, including mechanism, material, device, system, communication, processing, and power supply. Furthermore, cost reduction is imperative to

facilitate large-scale production, aligning with industry standards for economic viability. To foster innovation in HSs sensors, various strategies such as establishing robust testing protocols, enhancing durability and integration, leveraging computer-aided design tools, exploring environmentally sustainable production methods, and incorporating intelligent manufacturing techniques hold significant potential for advancement.

Despite present constraints, the progress in wearable devices for HSs monitoring holds significant promise for early detection of CVDs. To assure wearable HSs devices' accuracy, reliability, and clinical validity for diagnosis of CVDs, technological, regulatory, and clinical concerns must be addressed. The trajectory towards the widespread adoption of wearable devices for HSs monitoring is compelling and potentially transformative for cardiovascular care. This could lead to the timely identification and intervention of cardiac irregularities, thereby forestalling the development of more severe health complications and, potentially, preserving lives.

6.1 | Challenges and future outlooks

The basic parameters that should be critically considered for the future of intelligent and wearable HSs devices are presented in Figure 11. In terms of achieving seamless integration with the skin, the hard-soft integration technique remains the most promising approach for developing high-performance wearable electronics in the foreseeable future. Materials characterized by an exceptionally low modulus (measured in kilopascals) offer solutions to the mechanical challenges associated with integrating skin with active device components or supporting substrates, presenting a significant avenue for future exploration. The exploration of novel or combined materials holds the potential to enhance advancements in two-dimensional semiconductors and dielectrics. Furthermore, due to their superior carrier mobility, metal oxide semiconductors show promise in surpassing organics in flexible electronic circuits. However, integrating novel materials may introduce changes in skin/device impedance, alterations in the skin's thermal parameters, electromagnetic effects on the skin, or potential biocompatibility challenges. Therefore, before practical implementation, researchers must thoroughly analyze the fundamental characteristics of the system and their implications for skin interaction.

Recent advances in materials science engineering have paved the way for the use of MOFs and MXenes to further improve wearable HSs monitoring devices. For instance, MOFs are recognized for their high surface area and porosity, which can be tailored to improve sensor

sensitivity and specificity.^{203,204} That is critical for detecting subtle HSs amidst physiological noise. Similarly, MXenes, a group of two-dimensional transition metal carbides, nitrides, and carbonitrides, exhibit excellent conductivity and mechanical flexibility, making them ideal for the fabrication of flexible and responsive sensors.^{205,206} The high electrical conductivity and controllable interlayer spacing of MXenes make them suitable for various sensor applications, including pressure and epidermal sensors.^{207,208} The integration of MXenes with other materials has led to innovations in sensor platforms. For example, MXene-based hydrogels have emerged as a promising option for epidermal sensors due to their softness and unique electrical properties.²⁰⁹ In addition, MXene-coated sensors have been used to monitor lithium dynamics and identify emerging pollutants, highlighting the versatility of MXene in sensor technology.²¹⁰ By incorporating these highly functional materials into innovative sensor designs, performance can be improved, and new areas of noninvasive cardiovascular monitoring can also be explored. Therefore, incorporating MOFs and MXenes into sensor technology has shown great potential and has been extensively studied for wearable sensors, with research highlighting their advancements in flexible sensor technology, spanning wearable devices and smart medical devices.²¹¹

Improving the selectivity of hyperspectral sensors for precise target identification amidst complex environments poses a significant challenge. Focusing on sensitivity, specificity, and response time is crucial to optimize wearable hyperspectral sensor performance. This can be achieved by optimizing sensor size and design to enhance sensitivity and adopting optimal materials and production procedures to improve specificity and response time.

The challenge of stretchable and bendable sensing systems in wearable electronics arises from mechanical pressures induced by human body movement. This leads to component displacement and motion artifacts that significantly affect measurement accuracy. In the future, integrating wearable hyperspectral sensors with various sensor types and systems holds promise. For instance, cooperative motion sensors could enhance wearable health and safety sensors, with algorithms adjusting signals to analyze, summarize, and predict motion velocity and acceleration impacts.

The complexity of distinguishing hyperspectral signatures amidst multiple simultaneous sounds underscores the necessity of noise filtering despite its inherent challenges. Additionally, the amalgamation of concurrent sounds can hinder their discernibility. Ongoing advancements in deep learning algorithms promise to refine noise filtration, facilitating improved differentiation among concurrent noise qualities.

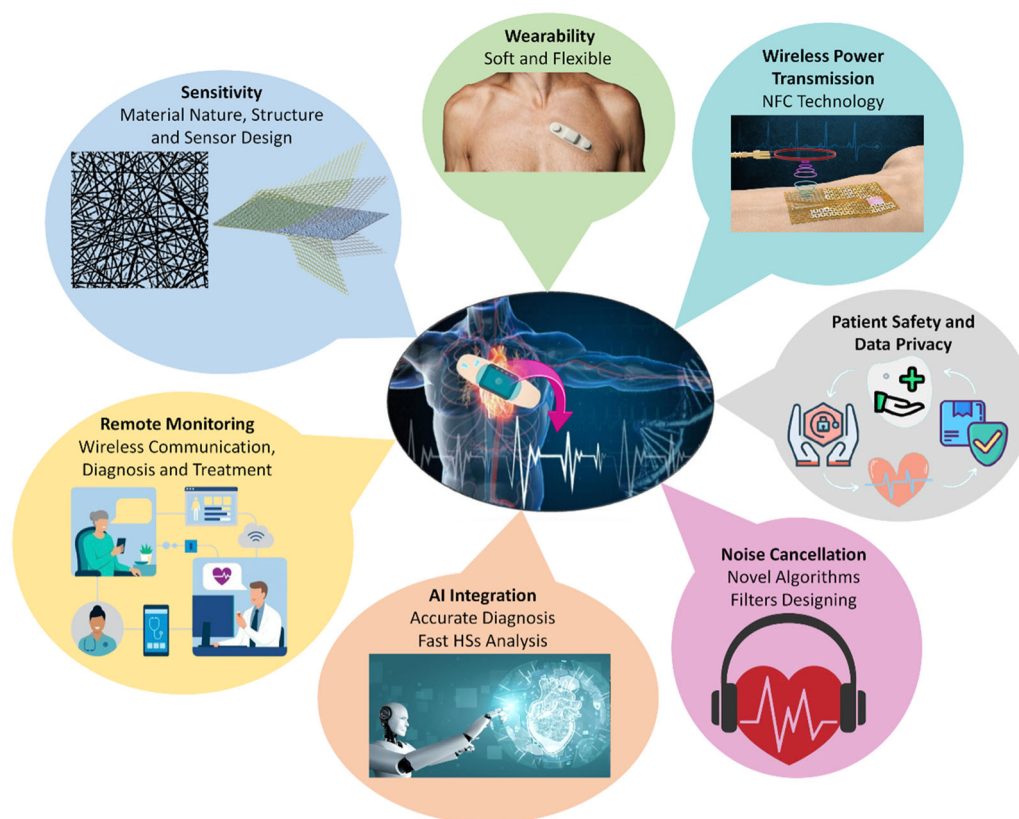


FIGURE 11 Parameters for designing smart and intelligent wearable HSs devices for the remote monitoring of CVDs. CVD, cardiovascular diseases.

Continuous electrical power is necessary for the wearable sensors to monitor human health remotely. Therefore, in addition to the sensor's sensitivity and reliability, integrating self-powered technology (battery-free) into wearable HSs devices can enhance and expand their functionality. As the next generation of IoT sensors, IoT sensors will employ various energy harvesting methods to develop a smart healthcare system. The necessity for self-powered wearable HSs sensors has become increasingly evident due to the rise in population, increasing rates of CVDs, and the growing number of older individuals receiving medical treatment. These sensors can continuously monitor patients in a closed-loop system by collecting the required physiological data to monitor the patient's health continuously. The benefits of smart healthcare based on IoT sensors include remote medical care, enhanced patient safety, improved treatment quality, and time and cost savings.⁷⁵

Advancements in wireless power transmission and communication technologies promise to enhance future HSs sensing systems from engineering and technological perspectives. The aim is to improve wireless power transfer and communication, enabling real-time monitoring capabilities and extending mobility beyond

clinical settings. Hence, future epidermal devices must incorporate sophisticated technologies for efficient energy transfer, transformation, and storage. One practical approach involves utilizing inductive resonant energy transfer for distant powering and recharging, a technique already prevalent in wireless recharging of cellular phones. NFC antennas offer the potential for simultaneous inductive resonant energy transfer for powering and inductive coupling for data communication, eliminating the need for batteries. However, addressing challenges related to the distance between transmitter and receiver and sampling frequency is imperative before implementation.

To advance the HSs sensing technology, fostering multidisciplinary collaboration with cutting-edge technologies like AI and IoT is imperative. Integrating HSs sensors with AI algorithms enhances robots' perceptual capabilities, enabling them to navigate complex environments effectively. Likewise, merging health sensors with IoT in healthcare facilitates real-time data storage, management, and analysis while integrating high-speed sensors into intelligent transportation systems. IoT and AI enhance communication between vehicles and their environment, promoting safer and more efficient transportation.

In the era of vast data utilization in scientific research and healthcare management, ensuring data security and privacy has become paramount. The emergence of data desensitization technology offers a critical means to mitigate risks such as unauthorized data disclosure and privacy breaches. Techniques like authentication and end-to-end encryption bolster communication security between patients and healthcare providers, safeguarding sensitive information effectively.

Rigorous clinical testing and validation are indispensable for establishing the viability of HSs device applications. Collaborative efforts among medical science, clinical medicine, and engineering experts are vital for the practical deployment of HSs sensing systems. Proof-of-concept studies are imperative to benchmark novel devices against established clinical gold standards, particularly for niche diagnostic applications leveraging innovative sensing mechanisms. Furthermore, extensive clinical trials are necessary to establish initial benchmarks and validate the accuracy and reliability of these systems.

The future of wearable HSs monitoring devices appears promising. Sustained research efforts, technical advancements, and interdisciplinary cooperation will drive innovation within this domain. Wearable technology has taken proactive steps toward reshaping cardiovascular care by addressing the considerations outlined. This has empowered individuals to take an aggressive stance in managing their cardiovascular health and has concurrently augmented the efficacy of healthcare systems.

7 | CONCLUSION

This review comprehensively examines recent strides in noninvasive, wearable, and flexible devices designed to monitor HSs continuously. The discussion encompasses various wearable technologies leveraging commercially available accelerometers, microphones, MEMS acoustic sensors, and flexible polymer-based patches. Notably, these technologies offer the advantage of compactness for clothing integration. These solutions present a practical and personalized approach to early cardiac disease detection by enabling real-time recording and analysis of HSs.

Nevertheless, this review acknowledges the challenges and limitations of wearable HSs monitoring technologies. Issues encompassing accuracy, data analysis, durability, and the need for robust large-scale experimental trials are among the highlighted concerns. These challenges underscore the ongoing necessity for research and development efforts to fortify wearable technology's overall efficacy.

In essence, this review underscores how wearable and adaptable HSs monitoring technology holds the potential to revolutionize the assessment of our

cardiovascular well-being. Continued research and development endeavors are imperative for surmounting current technological and practical constraints, ultimately refining accuracy, durability, and user acceptance. Wearable technologies focused on HSs monitoring stand poised to significantly impact the early detection and management of cardiac disorders, thereby enhancing patient outcomes and quality of life—an evolution driven by relentless technological advancement and innovation.

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CONFLICTS OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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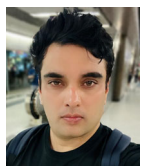
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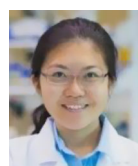
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